

AN AFRICAN GOLD NOVEL
THE SPIRITUAL FOURTH

AFRIKA 2100

They were never here for the gold.

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ARJUNA BADGER PRESS

Witwatersrand, west of Randfontein • Landing Day, 2071

She was at the third benchmark by half past four, which was earlier than the job needed and later than she had wanted.

The brass plug was where it had been in 2069, set into a concrete block the size of a bread loaf, stamped **TR/W-114**. Dineo brushed the grit off with her thumb, seated the tripod, and levelled up. Tebogo held the staff at the backsight and complained about the cold in two languages.

“You could have taken leave,” she said.

“Everybody took leave. There’s nobody left to take leave from.”

“Then hold it plumb.”

Overhead the object had not moved in any way a person standing still could measure. It had been up there since she was forty-four. Her sister had cried about it for a week in 2066 and then stopped, and now made jokes about it at Christmas, and this was the general history of the country’s feeling on the matter.

Closure on the morning traverse came out at eleven millimetres over 2.4 kilometres. Acceptable. She wrote it in the field book in pencil, because ink runs and pencil doesn’t, and because her supervisor in 2041 had rapped her knuckles for using a pen and she had never once forgiven him or gone back to ink.

The gravimeter went in at the fourth station, on the pad she had

poured herself in April. Base reading logged 05:12. She would take it again every hour until the thing was on the ground and every hour after that until dark. The Department wanted to know whether the reef moved. Two hundred metres of old stope work lay under this veld, mined out, flooded, and abandoned before her grandmother was born, and if you set anything heavy on the roof of that you might find out what a sinkhole thinks. Nobody in the country had been able to say what the object weighed. The estimates on the radio ran from four hundred tonnes to nine thousand.

“They should have landed it on the Free State,” Tebogo said. “Flat. Solid. Boring.”

“They didn’t ask me.”

“They asked somebody.”

They had, in fact, asked. Coordinates had come down through the mesh in March, in three formats, one of them a format nobody had used since 2038, and the Department had spent five weeks arguing about whether to counter-propose. In the end the Minerals people looked at the old plans and said the ground would hold and the Department said fine, and a fence went up, and Dineo poured her pads.

By seven the radio in the bakkie had gone over to the outside broadcast and she could hear the crowd on the ridge behind her without turning round. Chairs. Somebody’s speaker. A boy shouting the same word over and over for the pleasure of hearing it come back off the mine dump.

The station had put a cricket commentator on it. This turned out to be the correct decision, because he was the only man in Johannesburg trained to describe something arriving slowly for six hours without lying about the pace. He described the light on the underside. He described the fence. He described a woman selling boiled eggs at the cordon and gave her name and the price of the eggs, which had gone up.

“Now, if you’re joining us in the North,” he said, “I’m told you’ve got sound only, so let me do this properly for you. It is a clear morning here on the reef. Wind from the southwest, light. The object is

presently at, they tell me, eleven thousand metres and coming down at about the speed of a lift in a government building.”

Then a professor came on and said the descent was consistent with the profile published in 2066, and had been consistent every day for five years, and that the interesting question was not the descent.

Then a man from Krugersdorp phoned in to read numbers.

He read them well. He had the published telemetry in front of him and he read the ephemeris the way other men read football results: the object’s track back through 2070, 2069, 2068, and Diri’s tapes before that, the whole beautiful boring line of it, and he got as far as the burn.

“August 2061,” he said. “That’s the deceleration onset off the tapes. Peer-reviewed, twice. I’ll say it again because nobody says it. August 2061.”

“And the note was sung in June 2064,” said the presenter warmly, “so we know exactly why they’re here, and I’ll tell you what, my brother, when I was a boy nobody sang for me like that.”

The caller said something about the numbers. The presenter thanked him for the numbers. The next caller wanted to talk about whether it was a public holiday.

Dineo took the 07:12 gravity reading and wrote it down.

The taxis stopped at about half past nine. Nothing announced it. She noticed the sound of the R28 change and looked up and the traffic on it was cars only, and then not even cars. Government instructions had achieved nothing all week. The rank achieved it in an hour, because the drivers wanted to see, and once the drivers wanted to see, the country stopped.

Her nephew Sipho arrived on foot at ten with three litres of water, a bag of oranges, and the news that his mother had bought out the spaza.

“Everything?”

“Mealie meal, tinned pilchards, candles, batteries.” He counted on his fingers. “Sugar.”

“She remembers ’64.”

“She remembers ’65,” he said, which was the worse year, and put the oranges in the shade under the bakkie.

He stayed. He was nineteen and had never known the sky without it. She put him on the levelling staff so that Tebogo could go and eat, and he held it straight, and he watched the object the entire time over the top of the staff, and did not once let it wander.

At 11:04 she took the reading. Nothing. The gravimeter is a drift instrument and its drift that morning was ordinary and dull, and she noted the drift, and she noted the temperature, and she noted that the object’s shadow had reached the fence line on the western side.

It set down at 13:26.

There was no wind from it. That was the thing she had not expected and the thing she would be asked about for the rest of her life. No dust. It came the last thirty metres slower than the last hundred, and the veld under it did not move, and the grass did not lie down, and then it was on the ground with its mass on the ground, and a noise came across the flat to her that she would put in the report as a *low tone, approximately 40 Hz, duration under two seconds, cause undetermined* and which sounded to her, at the time, like a large door in a big empty building being closed by someone being careful.

The crowd on the ridge made no sound for a moment. Then it made an enormous sound, and then it went back to talking.

Dineo watched the gravimeter.

Eleven microgal. Then eight. Then, over four minutes, it settled and sat down at a new number and stayed there. Tilt at the two pads: 3.1 and 2.6 arc seconds, west-down, no rate. The old stopes had taken it. She had expected them to and had not been sure, and she wrote all of it down, station by station, in order, with the times, in pencil.

“Auntie,” Sipho said. “Is that it?”

“That’s it.”

“Should I put the staff down?”

“Put the staff down. Come and eat.”

Behind them a woman was crying and laughing at once and telling someone on the phone to be quiet and just *listen*, and a man with a portable set had turned it up so that the ridge could hear the studio, where a panel booked five years ago had finally run out of things to say and was reading messages instead.

Sipho ate an orange and said, “They came here.”

“Mm.”

“Not to America. Not to China. Here.” He was grinning. He was not making an argument; he was younger than the approach and he had grown up under it and it had come down on his ground, in his lifetime, on a Thursday. “That has to mean something about us, Auntie. Hey? They chose. That means something about *us*.”

Old Ntate Molefi, two chairs along, said, “It means the coordinates were good.”

“It means something about us,” Sipho said again, happily, and nobody on that ridge argued him down, and the woman with the phone said *amen* absently, still listening to the studio, and somebody further along said that G should have seen this, and somebody else said he did see it, meaning the arithmetic, and both of them were right and neither of them cared to fight about it.

Dineo took the 14:00 reading.

The delegates went in at four.

They had been at the Rand Show grounds for three days in a marquee, arguing about who would speak first, and in the end the order was alphabetical by delegation in French, which pleased nobody and was therefore adopted. The radio carried all of it because the radio carried everything that day. Dineo listened while she re-ran the levelling loop, because the ground had moved and the ground now needed measuring again, and because a person can only look at a thing for so long.

They asked about weapons. Answer: a statement of carried systems by category, three items, with tolerances, and a declared refusal to

discuss capability.

They asked about origin. Answer: a bearing, an approximate transit duration, and a refusal on the location.

They asked whether we are alone. Answer: no.

The ridge went up like a stadium at that and Dineo missed the next four questions.

They asked whether there is a god. Long pause on the tape, which the whole world afterwards agreed was the most human moment of the day and which was, on the recording, four and a half seconds of room noise. Answer: *the question is not one we are competent to hold.*

They asked intentions. Answer: a manifest. A list of quantities of worked gold at stated purities, a schedule, and the word *renewal*, which was in the second sentence and which the translators fought over for a decade.

They asked, again and again, in six languages, whether the note had brought them, and the answer each time was at protocol level and did not say yes, and every delegation in the marquee wrote down yes.

A delegate from the East African bloc asked if they had a name for themselves. Answer: a designation, eleven characters, given twice.

By six o'clock people in the marquee had begun to say *Anunnaki* out loud, first as a joke and then not as a joke, because it was the oldest word anybody had for people who come a long way for gold to mend a dying thing, and the fit made the hair stand up on your arms. The visitors did not accept it. They did not correct it either. They let it sit on the table like a hat somebody else had put there.

Nobody asked what they needed.

Dineo noticed, in the way you notice a missing bolt, and said so to Tebogo while she packed the prism: that when a client comes onto your site the first thing you ask is what he needs and the second thing is what it will cost to do it properly, and that they had done neither in nine hours.

"They asked about weapons," Tebogo said.

"They asked about weapons."

“Well.” He shut the case. “The meat won’t cook itself.”

He was quoting somebody’s uncle from the morning phone-in, and it was already the funniest sentence in the country, and it stayed the funniest sentence in the country for seventy years.

The braai smoke came up off the ridge at dusk and the taxis started running again at about four the next morning, which is to say the day ended the way days end.

Dineo sat in the bakkie with the door open and the field book on her knee and the reading lamp on and worked through the day’s stations in order, transferring them onto the Department’s forms in her small upright hand: TR/W-114 through the loop and back, times, temperatures, closure, the eleven microgal, the tilt at both pads, the 40 Hz tone with *cause undetermined*, the new datum. All of it true. She initialled each page.

Before dawn that morning, an hour before she seated the tripod at the third benchmark, she had walked out past the fence line to a place that was not a station.

Her pencil stopped above the next ruled line.

The lamp buzzed. On the ridge somebody’s speaker was playing music now, and the crowd was thinning, and Sipho was asleep in the tray with his jacket over his face.

She looked at the line for a while. Then she took the pencil away, closed the field book on it, and put the book in the canvas bag and the bag under the seat, and she did not write it down, then or afterwards, for twenty-seven years.

The forms went in on Monday.

By Tuesday she had four requests: a site plan at 1:500 for the cordon, a levelling datum the delegations could all agree to use, a footprint boundary to three decimals for the ledger, and a note from a man in Pretoria asking whether the Department’s monument stamps could be re-cut, since TR/W-114 now sat inside a restricted area and the numbering would have to change.

She re-cut the numbering. It took eleven days.

8WITWATERSRAND, WEST OF RANDFONTEIN · LANDING DAY, 2071

eGoli Spaceport, Berthing Floor Four • six weeks to Renewal

The first thing Naledi did on shift was mistrust her own ear.

The reference bar lived in a steel case bolted to the wall outside the choir line, and the case had a temperature log, and the log had to be initialled, and the bar had to be struck with the wooden mallet and not with a knuckle, no matter how clever your knuckle. She struck it. She held the earpiece off, listened raw, wrote *A-440 ok by ear* in the book, then put the earpiece on and let the counter tell her she was eleven millihertz optimistic.

Eleven. She wrote that down too. Some mornings she was flat, most mornings she was sharp, and in nine years she had never once been right, which was the whole reason for the mallet and the case and the log.

"You're sharp," said Thandeka, coming past with two coffees and giving her neither.

"I'm sharp before eight. After eight I'm a saint."

"You're sharp at fifty. My mother was sharp at eighty. It's genetic and it's ugly."

The floor smelled the way it always smelled: hot platcap bank, cold rock, machine oil, and the sweetish burnt note off the gantry brakes that everybody claimed to hate and nobody ever complained about in writing. Under the smell was the noise, which was not noise once you

had lived in it. Floor Four hummed. Gallery air came up through the grating at eighteen degrees and made a broad grey sound in the low sixties. The e-grav hoists over Bay Two sang thin and high while their rotors were up, and dropped out of hearing when the freight settled. Footfall pavers clicked under everybody's boots, harvesting whatever a person's weight was worth, which was almost nothing, which was the point. Waste no motion. It was written on the wall in the concourse in three languages and a man had been paid too much money to design the lettering.

Naledi loved it here in a way she had given up trying to explain to people at parties.

The choir line was nine stalls in a row against the west wall, each one a lead-lined box the size of a shower, each with its driver ring, its pickups, its little brass door with a proper handle because the guild's founders had come out of harvester engineering and did not trust anything that opened by itself.

Worked gold went in. A reading came out. That was the job, and it took nine years to learn and about forty seconds to describe.

Lot seven of the Kenyan consignment was already sitting in Stall Three under custody seal, thirty-one kilograms, formed to what the manifest called a Class-B ring blank and what everybody on the floor called a doughnut. Naledi broke the seal in front of the Steward on duty, because that was the order — Steward first, engineer second, ledger third — and the Steward wrote the time.

Tefo, that morning. He was her age and he stood like a much older man.

"Morning, ntate."

"Morning." He watched her hands. He always watched her hands. She had decided years ago that this was not suspicion but a kind of attendance, the way her mother watched a pot.

She seated the doughnut on the spider, dogged it down finger-tight and then a quarter turn, and ran the sweep. Twelve seconds of rising drive, the pickups reading back, the counter building its picture. The stall answered. It always sounded to her like a struck glass held under

water: not a note, a *place* where notes went.

Fundamental within four millihertz of the Accord figure. Second and third partials where they should be. Q high. The decay curve came up on the screen and lay down over the reference envelope like a hand into a glove.

“Clean,” she said. “Beautiful, actually. Whoever formed this one wasn’t rushing.”

“Nairobi,” Tefo said.

“Well. Tell Nairobi.” She keyed the sign-off, thumb on the plate, and the ledger took it and gave her back a receipt hash she would never look at and could never alter. *Put it on the ledger and it’s out of your hands* — half the guild’s manuals opened with a line like that, and half the guild recited them like grace before food, which Baba Siphosaid was fine as long as they also read the appendices.

Lot eight failed.

Not dramatically. Eight was a plate, eleven kilos, and its fundamental sat nineteen millihertz low with a partial smeared out into a fat little hill that had no business being there. Naledi ran it twice, changed the seating, ran it again, and got the same hill.

Karabo, who was two months out of the academy and would be excellent in about four years, leaned in and said, “Contamination?”

“Sometimes.”

“Silver?”

“Sometimes it’s silver, sometimes it’s a void, sometimes the tolerance on the forming press walked in the night and nobody told anybody.” She flagged it, wrote *reject, partial structure at 2f, recommend section and re-melt*, and let the stall wind down. “Don’t guess in the box. Guess in the canteen.”

“So what is it?”

“I think it’s a void.” She shrugged. “In the canteen.”

Tefo wrote the time on the reject and did not say anything, and the plate went back into custody, and somewhere in Kenya a foreman was going to have a bad Tuesday. That was the trade. The visitors

bought worked gold at stated purity in stated forms to stated tolerances, and they did not haggle, and they did not accept substitutes, and in twenty-seven years no consignment had ever been refused at the pad, because the guild refused it here first. Baba Sipho put it plainly at every induction: *we are the last people who can still say no cheaply.*

The hull for Bay One came in at 11:40, which was 11:40, because it always was.

Naledi went up on the catwalk to watch, along with everyone else who could invent a reason. The gallery mouth is a vitrified bore forty metres across, glassed by something nobody has explained in seventy years of trying, and the catwalk runs straight through it, and she had walked that catwalk so many times that she only sometimes remembered to be amazed. The hull came over the apron without wind. She had heard her grandmother describe that once, on the radio programme they replayed every Landing Day, in the voice of a woman being asked about weather.

Protocol came through on the floor board in the usual form.

BERTH ONE. MASS AT CONTACT 3,410 TONNES. CLEARANCE REQUESTED 4 METRES. HOLD 6 HOURS 40 MINUTES. UPLIFT LOT SCHEDULE 44-C, NINE ITEMS, PURITY AS FILED, FORM AS FILED. TOLERANCE UNCHANGED.

“Tolerance unchanged,” Thandeka read out. “Every time. As if we’d wake up one morning and start rounding.”

“They’re a freight company,” Naledi said. “You’ve met freight companies.”

“I’ve met freight companies who send flowers.”

The liaison’s countersign came a minute later, six lines, signed *Sela*, and one of the lines noted that item four of 44-C had been re-formed since filing and asked that the tolerance be verified against the filed figure rather than the re-formed drawing. Which was correct. Which was, in fact, exactly what Naledi would have asked for in their position, and she had a small private habit of noting when the counterparty was being careful, because on a floor where everybody talked about the

visitors as an event, it was steadying to remember they were also a client with a paperwork culture.

Karabo, chin on the rail: "Do you ever think they're watching us work?"

"They're watching the load," Naledi said. "Which is us working."

She ate outside, on the wall by the east cordon, because the canteen had the sound of eighty people and no ceiling treatment and after five hours of pickups her ears wanted grass.

The fence there is chain-link, four metres, with the pilgrim gate at the end of it, and by noon there were maybe sixty people along it. Ordinary. Some had come for the hull; you could see the hull's upper curve from that stretch, and people held children up. Some had come for the other thing. A woman was pressing a coin against the wire and holding it there with her palm flat, eyes shut, and further down two men had chalked the sentence on the pavement and were arguing about whether the short form counted.

The coins were the ledger coins, the ones with G's face on the obverse, worn smooth around the jaw from exactly this. Naledi had one in a drawer at home from her confirmation, which had been a Fold confirmation on her father's side and a very tense lunch on her mother's.

A young man came along the fence with a clipboard, doing it politely, and asked her whether she worked inside.

"I do."

"Then can I ask you," he said, "what would he have signed?"

He meant the Renewal. He meant it seriously; he had a lanyard and a printed sheet and the patience of somebody who had asked forty people already.

"I don't know," Naledi said. "He died before the Flare. He never saw any of it."

"He saw the arithmetic."

"He saw some of the arithmetic." She wrapped the rest of her bread. "Ask me what the tolerance on item four is. I'll tell you to three places

and you'll be bored."

He wrote something down anyway. She hoped it was *bored*.

Her mother left a voice note at two, which the mesh delivered in a burst when Naledi came out of Stall Six.

Ausi, the man from the Department phoned again about Gogo's field books. I told him the family has them and the family is not a library. Phone your brother, he thinks I was rude.

Fourteen books. Canvas-bound, pencil throughout, initialled at the foot of every page in a small upright hand, and in the margins of four of them a set of bearings and offsets that did not correspond to any monument in the register. Naledi had photographed those pages when she was twenty-three, for reasons she could not now reconstruct, and never shown them to anybody at the port, for reasons she also could not reconstruct.

Gogo Dineo had surveyed the landing ground. Everybody knew that. Every Landing Day somebody played the tape. And she had sat in her chair in Qwaqwa with her ankles swollen and answered every question a granddaughter could think of about datums, closure, and the correct way to hold a staff, and none at all about anything else, and she died in 2098 with her mouth shut and her handwriting intact.

Naledi put the earpiece back in and went to lot four.

Handover at seventeen hundred.

She read the numbers out to Lebo off the book and not off the screen, because that was the discipline: *lots seven through fifteen signed, lot eight rejected for partial structure at 2f, stall three driver ring at nine hundred hours since service, stall six pickup two reading two millihertz proud all afternoon and I've noted it, floor idle logged at fourteen-hundred, uplift 44-C staged, item four verified against filed.*

"Pickup two proud," Lebo said. "Proud how?"

"Proud like a cousin with a new car. Two millihertz. It's in the book."

"You want it changed out?"

“I want it watched.” She signed. Lebo signed. The ledger took both and gave nothing back.

She hung the earpiece on its hook, put the mallet in its clip, initialled the temperature log for the reference bar on the way out, and stood for a second in the doorway with her jacket half on.

The floor was doing what it does in the evening: gallery air coming up grey and broad, one hoist still spinning down over Bay Two, a hull sitting on its pad with its mass on the ground and the room’s idle sitting under all of it, the sound she had worked inside since she was nineteen.

She hummed against it, the way you touch a wall to see if it’s wet.

Then she went up the stairs and out through the concourse where the pavers clicked, and got on the tram, and gave up her seat to a man with a cooler box, and rode home with her fingers going on her knee, still holding the pitch she had hummed, trying it against the pitch she had walked out with. Nothing she could put a number to. Not enough for the book.

Sharp before eight, she thought. Sharp at five, apparently. Genetic and ugly.

She let it go twice on the way home and picked it up again both times.

Lancaster Compact zone, Gordonville depot • six weeks to Renewal

The scale said 908 kilograms and the manifest said 908 kilograms, so Ruth signed the manifest.

That was the third weighing. First at the mint in Philadelphia, second at the rail transfer in Coatesville, third here on the depot platform scale that the Compact used for hog lots and hay and now for this. She had asked for the third weighing herself. Nobody argued. A quartermaster who signs on two weighings is a quartermaster explaining herself to a board in six months.

Twenty-two cases. Each case forty-one kilos of worked gold and eleven of crate. Ash crates, dovetailed, no nails in the corners, made in Intercourse by a shop that mostly built grain wagons. Steel banding. Custody seals on the bands, wire-and-lead, the old kind, because the Accord wanted a seal a man could look at.

She walked the row and put her hand flat on each lid. Not for anything. It is what you do.

“Load’s good, Captain,” Beiler said.

“Load’s good. Wagon.”

They went and looked at the wagon.

It was a gun carriage. It had been a gun carriage in 2066 when the

Army moved field pieces out of Carlisle to sit on the granary roads, and the barrel had gone for scrap in the seventies, and the trail and the axle and the wheels had stayed because nothing built for a gun ever wears out doing anything else. The shop at New Holland had decked it, cross-braced it, hung new sides, and fitted piezo struts in place of the leaf springs, so the ruts on the Blue Ball road would put something back into the cells under the seat.

Two teams of Belgians. Four horses. Sixteen hundred kilos of animal for nine hundred of cargo, which the young men found funny and which Ruth did not, because the animals were the part that would still be working if every cell in the county went flat.

"Reuben says the offside wheeler is short in the shoulder," Beiler said.

"Is he."

"He says it. He's been saying it a week."

"Change him out."

"Reuben doesn't want him changed out. He wants it on the record that he said it."

"Then put it on the ledger and keep the horse." Ruth crouched by the hub and turned the wheel a quarter and listened to the bearing. "Tell Reuben I put it on myself."

Beiler wrote it in his book. He wrote everything in a book. He was forty-four and had been Plain until he was twenty-two and was still Plain in every way that mattered to a load: he did not hurry, he did not shout, and he could tell you the weight of a thing by looking at it and be within five kilos.

The blessing was at seven, before the heat.

Two hundred people came, which was more than she wanted and fewer than she had feared. The depot yard filled up along the fence, and the school let out for it, and somebody's Assist frame stood at the back with its arms folded, and the man in it apologised twice for the noise of the joints, which was nothing, which was less noise than the children.

Bishop's daughter's husband led it. That was how it worked now; the Fold has no ordination and it keeps the shape of the households it grew out of. His name was Eli Hoover and he sold seed.

He stood on the platform scale, which Ruth noticed and liked, because it meant he was being weighed.

He read the folded passage about the corn of wheat. Then the one out of the second book of the Ledger, the ninth kindness, the one about the neighbour's ox and the neighbour's roof, which in the folded text has the ox and the roof and then the sentence, and the sentence is *be kind*.

She had heard him read it before with the whole line in it. Not this morning. Two hundred people and the school children in the front.

The gathering said it back the way a gathering does: half a beat behind, some in Deutsch, some not, the two languages sliding over each other and coming out even.

Sei gut. Be kind.

Ruth said it. She said it with her hat off and her hand on the wagon side, because the wagon was what they were blessing, and she meant it, and she also stood there with the other half of it in her mouth like a stone.

Eli Hoover put his hand on the near case.

"This is not a tribute," he said. "We don't know what it is. We're sending it because we said we would, and because they asked for the work and not the metal, and that's a thing we understand — a man who wants your work is dealing straight with you."

Somebody said amen.

"It might be payment," he said. "It might be a gift. Captain Stoltzfus is going to find out and come back and tell us, and we'll believe her, because she doesn't say things twice."

Ruth kept her eyes on the case bands. That was not the arrangement. The arrangement was that she was quartermaster of a consignment and would deliver it into the port's custody and obtain a receipt. But she did not correct a man in front of his own gathering over a load,

and she noted that she had not corrected him, and she knew she would think about that on the water.

Then the horses were prayed for by name. That is the Plain part and it has never left. Dan, Jake, Sadie, and the offside wheeler, whose name was Hosea and who was short in the shoulder.

The children sang. Two verses and a chorus, nothing anybody in Africa would recognise as a hymn — the tune had come from a hunger-year work song and everybody knew it and nobody knew who made it.

Her mother did not come. Her mother was three miles up the Musser road and eighty-one and had said in April that she would not stand in a yard and watch the Army bless gold.

Her brother did not come either. Levi was fifteen miles away and had been fifteen miles away for twenty-three years. He had stayed; she had gone. He farmed her father's ground and did it better than her father had. They wrote. He signed his letters *your brother Levi* and asked after her back.

She had gone up on Saturday and sat in the kitchen and eaten pie and been asked nothing about the consignment, and had answered nothing, and it had been a good afternoon.

At half past eight the yard emptied and she went round the wagon alone with the checklist on the clipboard, which was paper, because paper does not need a cell.

Bands: twenty-two, seals intact, numbers matched to the manifest. Chocks: in. Brake shoes: clean and dry. Struts: four, charging, needle moving. Cell bank under the seat: at sixty-one percent, which was fine, which was better than fine for a wagon that was going to spend nine days on Compact roads before it saw a railhead. Water: forty litres for the animals, tarpaulined. Feed: eleven bags oats, four bales. Escort: six, mounted, small arms, plus Beiler. Rifles: seven, cleaned last night in front of her. Bible: her grandmother's, German, in the flat pocket of her duffel, where it had been since Fort Lee.

She got to the bottom of the list and stood a minute in the shadow of

the wagon with the yard smelling of horse and dust and the platform scale ticking as it cooled.

She took the coin out of her breast pocket.

It was a ledger coin, worn to a soft brown on the high spots, his face on the obverse rubbed down mostly to the jaw and the eye socket. On the reverse the two sentences, small caps, all the way round the rim, the way the honest ones have them.

DON'T BE A CUNT. BE KIND.

She read it. She read it every time. There was no ceremony in it; it took two seconds; it was the same as putting a hand on a lid.

The soft form is a kindness to children and grandmothers and she had no argument with it. She had watched it happen in her own lifetime — the coins with the word filed off, then the coins struck without it, then the passages read short in company and long alone, and half the gatherings in the county now with the full line on the wall in the back room and *BE KIND* over the door.

It is a warning first. That was all. The first sentence tells you what you will do without trying, and the second tells you what will cost you something. Take the first one out and you are left with a nice idea and no floor under it.

She put the coin back in her pocket and buttoned the pocket.

She was not baptised in. She had been nine weeks from it at eighteen and had gone to the recruiter in Lancaster instead, and no one in her family had ever once made it into a wound in front of her, which was its own kind of pressure. She had a coin and no baptism and a German Bible she could read slower than she could shoot.

Beiler brought her the road order at nine, and the weather off the mesh, and a peach.

"Ninety-one today," he said. "Ninety-four Thursday."

"We'll water at Blue Ball and Gap."

"Reuben says Gap has the trough out for repair."

"Then Reuben knows where the next water is and he'll tell me at Blue Ball." She ate the peach. It was a good peach. "Amos."

"Captain."

"When we get there. If I'm asked to stand up and say what this is."

He waited. He was good at that.

"I don't know what it is," she said.

"No."

"Eli said I'd find out."

"Eli sells seed," Beiler said. "He also sells hope, and he doesn't charge for it, so it comes out about even." He put the peach stone in his pocket rather than the yard. "You'll say what you saw."

"That's the whole of it?"

"That's the whole of it. You've never been any good at the other."

She walked the row of cases one last time, twenty-two lids, hand flat on each.

Then she climbed up and took the seat on the left because Reuben drove and a quartermaster does not drive, and Reuben said something to the near team, low, in Deitsch, that was not an order and not a prayer.

The chocks came out. The wheelers took the weight; the leaders leaned into it; the trail creaked at the pin and the whole nine hundred kilos came off the platform apron and onto the gravel, and the needles on the strut gauges jumped as the first rut took the offside wheel.

Somebody's dog came out of the depot office and ran the length of the fence with them, barking, and gave up at the gate.

They went out the Gordonville road at four miles an hour, east into the sun, with the escort at a walk behind and the dust standing up in the ditch grass and not settling until they were past the schoolhouse.

eGoli Spaceport, Berthing Floor Four • six weeks to Renewal

The discrepancy came out of a housekeeping job nobody wanted.

Uplift 44-C had gone out on the Thursday hull and the guild's part of it was closed, but the Accord's annexes require the tuning house to reconcile each uplift against the counterparty's stated use, once, within seven days, and file the reconciliation. It is a paper exercise. The visitors file a repair manifest for every consignment: item, form, mass, and a use-line saying which component the metal is for. The guild files back a resonance sheet saying what each item measured. The two go into the ledger side by side and nobody reads them again, ever, in the whole history of the trade.

Naledi drew it because she had lost a bet about a rugby match.

She took the terminal in the tuning office, which is a converted store cupboard with a window onto the gallery mouth and a kettle that trips the ring main if anyone runs the desk lamp at the same time. She had the lamp off. She had 44-C's nine resonance sheets on the left of the screen and the manifest on the right, and a mug going cold at her elbow, and she was going to be finished by four.

Item one. Ring blank, 62.4 kg, use-line: *guide segment, ring assembly, class 3*. Sheet: fundamental 212.113 Hz, partials clean, Q high.

Item two. Plate. Use-line: *coupling face, secondary*. Sheet: 388.006.

Item three, four, five. Fine.

Item six was a ring blank too. Same class, same nominal mass within a kilo, same use-line word for word: *guide segment, ring assembly, class 3*.

Fundamental 212.098.

Naledi looked at that for a while.

Fifteen millihertz is nothing. Fifteen millihertz is her own ear before eight in the morning. Two ring blanks formed to the same drawing in two different shops on two different presses will differ by more than that, and the tolerance band in the annex is generous by design, because the visitors buy to a tolerance and not to a number, and the guild's whole existence is the width of that band.

But the guild does not read the band. The guild reads the *shape*.

She pulled up the two decay envelopes and put them one over the other. Item one's third partial came down like a dropped stone. Item six's third partial came down and then sat, for about four hundred milliseconds, on a shelf.

She had seen that shelf before. Everybody had. It is what a driven mass does when it is coupled to something that wants to keep ringing at a slightly different pitch — a stall with a loose spider, a plate that has been seated against the lead lining, a pickup two millihertz proud. It is a coupling artefact and the correct response is to reseal and rerun, and if it recurs, service the stall.

Item six had been read in Stall Five. Item one in Stall Five. Same stall, same morning, same engineer — Lebo, who was better at this than Naledi and had signed both without a note.

She got up, went out onto the floor, and reseated nothing, because both items were four hundred and eleven kilometres up and out.

So she did it the long way, which took nine days and mostly happened between other things.

She pulled every reconciliation for every consignment of the last three years. Two hundred and sixty-one uplifts, one thousand nine

hundred and forty items. The ledger holds them all and the ledger is not slow; the slow part is that the use-lines are free text.

The visitors write their manifests in protocol English and their protocol English is exact but it is not a database schema. *Guide segment, ring assembly, class 3. Guide segment (ring), class three. Ring assembly guide, cl.3, replacement.* Naledi spent two evenings building a table by hand to collapse the variants, and she built it in pencil first, on the back of a stall service log, because that was faster than arguing with software.

Then she sorted every item in the three years by use-line and looked at the resonance sheets grouped that way.

The class-3 guide segments came out in two families.

Not a spread. Two clumps, with a gap between them you could park a hull in. Family A sat around 212.113 and had clean fast decay. Family B sat fifteen to nineteen millihertz low, always low, never high, and every single one of them had the shelf on the third partial.

Two hundred and six class-3 guide segments in three years. Ninety-one in family B.

She sat back and the chair squeaked and she said, out loud, to the kettle, "Oh, that's not nice."

Because a use-line is a *use*. It says what the metal is for. And if the counterparty is buying two acoustically distinct components and filing both under one use-line, then either they have a sloppy clerk, or the two families go to two different places, and the manifest is not describing the second place.

She checked the sloppy clerk hypothesis, because you do. She sorted family B by filing date, by forming shop, by mass, by purity, by the shipping hull, by the day of the week. Family B came from Kenya and Australia and the Bushveld and Chile and one lot out of Aksum. Family B included items formed to the drawing by four presses that had never made anything else. Family B ran all three years with no trend, roughly two in five, and the split held to within a couple of percent across every quarter she cut.

Nobody's clerk is that consistent. Consistency is the fingerprint of a system.

She did the last check on the ninth evening with the lamp off and the kettle unplugged. She took the twelve family-B items with the most complete sheets and looked not at the fundamental but at the interval between family A and family B: 212.113 against 212.098, 388.006 against 387.978, 145.220 against 145.209.

She worked the ratios on paper. They came out the same to four places. Same interval, every pair, every form, three years. Family B was not a family of flawed parts. Family B was family A transposed — the whole set shifted down by one fixed, tiny, deliberate amount.

You do not get that from voids. You do not get that from silver. You get that from somebody forming metal to a *different reference pitch* and calling it the same component.

Naledi's stomach did the thing it does on a hoist.

She wrote it up properly. She had it in her hands by eleven that night and she did not send it, because at eleven at night everybody is a genius. She read it again at half past six with the reference bar log in front of her and her ear eleven millihertz optimistic as usual, and it still stood up.

Discrepancy report. Guild form 12. Observation, method, data appendix, and one recommendation: *that the tuning house request clarification from the liaison as to whether use-line "guide segment, ring assembly, class 3" covers two distinct component specifications, and if so, that the annex reconciliation format be amended to distinguish them.*

That was all she asked for. A clarification and a form change. She was rather pleased with how boring she had made it.

She filed it Tuesday, 08:14, and the ledger gave her back a receipt hash, and she went and did lots eleven through nineteen.

The file came back on Friday.

STATUS: CLOSED. DISPOSITION: NO ACTION REQUIRED. NOTE: VARIANCE WITHIN ANNEX TOLERANCE. RECONCILIATION FORMAT UNDER REVIEW BY COMMISSION AHEAD OF RENEWAL; SUBMISSIONS SUSPENDED.

Time between filing and closure: seventy-nine hours, which included a weekend.

She stood in the corridor outside Stall Two with her earpiece around her neck and read it three times and could not find the part that was wrong, which was the annoying bit. Every sentence was true. The variance was within tolerance. The format was under review; she'd had the circular in March and deleted it.

"Two millihertz proud," she said to nobody.

Karabo went past with a spider wrench and said, "Sorry?"

"Nothing. Get me the service book for Five."

Stall Five's driver ring had been changed in February. Pickups replaced last August, both. Calibration against the bar within four millihertz every week for fourteen months, initialled, dated, temperature logged. The stall was not sick. Naledi had known that before she asked. She asked because a person who files a report and then does not check her own instruments is a person who will be asked why she did not check her own instruments.

Marais came down to Floor Four on the Wednesday, which was not remarkable. He came down most months. He had come out of the guild himself, forty years ago, and he still put on the earpiece and asked to hear the idle, and the floor liked him for it and the floor was not wrong to.

He asked her to walk with him on the catwalk. He was seventy-one and he walked slowly through the bore and stopped at the middle, where the glass gets that green depth in it, and put his hand on the rail.

"You know what my father did on this apron," he said. "He drove a water truck. Nineteen years."

"Ntate, I know."

"I tell everyone. It's the only story I have." He looked down the length of the bore. "Your report was good. I want to say that first, and I want you to believe I mean it, because in about four minutes you're going to think I'm managing you. It was good. The interval work especially

— you didn't chase the millihertz, you chased the ratio. That's the guild at its best and I read it twice."

"Thank you."

"It's also correct, as far as I can tell." He said it plainly, the way you'd read out a mass at contact. "I don't know what those ninety-one items are. Neither do you. Neither, I suspect, does anyone in this building. Now let me tell you what week it is."

He let go of the rail and turned to face her.

"Six weeks. The Renewal instrument is a hundred and nine pages and the volume schedule alone took four years to agree, because volume is the only clause in it that the other twenty-two signatory blocs actually fight over. China wants doubled tonnage and a shortened cycle. The Southern Carry wants the cycle held and the technique schedule front-loaded. The North American Compact wants proof it isn't paying rent. And every one of them has domestic weather that would eat their delegation alive if they came home with less than they promised."

"I know."

"You know it the way you know the weather in Chile. Let me be specific, because specific is the only thing you and I both trust." He counted on his fingers, no theatre in it. "The platinum framework is annexed to the Renewal. Not adjacent. Annexed. If the Accord lapses without instrument, forty percent of the world's platcap electrode supply falls back to bilateral contracts negotiated in a panic, and the price of a cell in Nairobi triples inside a quarter, and every mesh node in the Congo basin that runs on a two-hundred-rand cell goes dark by degrees. Second. The technique schedule. There are eleven deliveries queued behind signature, one of which is a ceramic process for piezo harvesters that halves the cost of a footfall paver. That is not glamour. That is a village in Limpopo lighting a clinic. Third, and this is the one I lose sleep on. This country has been indispensable for twenty-nine years. Do you know how long any country has ever been indispensable?"

"No."

"Neither does anyone, because nobody has been. We are hosting the species relationship on our ground, under our custody, on our terms,

and we are doing it because the Accord says so. Instruments say so. Take the instrument away and what's left is that we happen to be sitting on the gold, and the world has a long and well-documented history of arrangements for people sitting on gold." He put his hands in his pockets. "My father drove a water truck on an apron owned by other people. I would like to die before that comes back."

A hoist came up over Bay Two and the rotors sang thin and then dropped away.

"So," Naledi said. "Drop it."

"No. Hold it." He shook his head slightly. "I'm not asking you to unsee it. I'd fire you if you could. I'm asking you to be curious in September. Curiosity has a season, Naledi, and the season is the twenty-seven years we've had and the fifty we'll have after signature. We have been allowed to be curious for twenty-seven years. We get to be sovereign for six weeks. If I take a question about ninety-one guide segments onto that floor now, I don't get an answer. I get a delegation from the Give bloc saying the host nation is inventing doubt to raise the price, and a delegation from the Keep bloc saying they *told us* the metal was going somewhere unstated, and eleven technique deliveries in a drawer, and the clinic in Limpopo not lit. And I still don't have an answer, because the counterparty will file six lines and sign them and the six lines will be true."

"They usually are."

"They usually are." Something moved at the corner of his mouth. "Re-open it in September. I'll co-sign the request myself, and I'll put my name above yours on it so it isn't a junior engineer against a protocol, and I'll make sure the format review takes your amendment. That's not a bribe. That's a plan, and it's a better plan than the one where you're right in August and unemployed in October."

He waited. He was good at waiting; he'd been in rooms.

"Understood," Naledi said.

"That's not agreement."

"No, ntate. It's understanding. You asked for one and I've got it."

He laughed, actually laughed, and patted the rail twice, and went off

down the catwalk with his hand out for the rail every third step, because the floor grating is open there and he was seventy-one.

She stayed in the bore a while.

She could not tell him the real thing, which was smaller than the country and larger than a form change. Lebo had signed both sheets. Lebo, who had eighteen years on her and had taught her how to hold the mallet, had put her thumb on the plate twice in one morning and certified two items as the same component, and they were not the same component, and Naledi had let a receipt hash go into the ledger under her own thumb ninety-one times over three years without seeing it.

She had written *A-440 ok by ear* every morning of her working life and then let a machine correct her, on purpose, because she did not trust her ear, and the whole point of that discipline — the mallet, the case, the temperature log, the humiliation at eleven millihertz — was so that nothing would ever get past the house. Not *nothing important*. Nothing.

Ninety-one items had got past. Twice a week, for three years, past everybody, into the ledger, out through the bore, up.

The gallery air came up grey and broad through the grating.

She went down to the tuning office and reopened her pencil table, and started a second column, and headed it with the interval — the actual number, four places — because if family B was formed to another reference, then somewhere there was a bar somebody had struck to get it, and she was going to find out whose mallet.

Brixton, her mother's house

• five weeks to Renewal

The survey books were in the second bedroom, in a tea chest, under a duvet cover nobody had thrown out.

Nine books. Buff card covers, cloth spines, the kind the Council issued until about 2080 and then stopped issuing, so Gogo had bought the last four out of a stationer in Fordsburg and complained about the paper for eighteen years.

Naledi's mother brought her tea and stood in the doorway with her arms folded.

"You're looking for what."

"Ground. There's a gallery mouth in the north-west quadrant that the port's plan says was capped in '74 and the walk-through says was capped in '81."

"And your grandmother is going to settle it."

"Her books ran from '69. Nobody else's do."

Her mother made a sound and went back to the kitchen. She had spent two years not sorting this room and was not going to be hurried into it by a daughter with a question.

The books are not diaries. Nobody who has read one would call them that.

Left page: station number, coordinates, instrument height, date, temperature, the initials of whoever held the staff. Right page: the sketch,

and the sketch is to scale, and where it is not to scale it says so, in capitals, with the fudge factor written next to it. Book Three had a whole leaf of tension corrections for a chain that had been dropped down a shaft and straightened by a man in Roodepoort who charged too much. Gogo had written the man's name and what he charged.

Naledi worked through them at the desk with the window open. Traffic on Fulham, someone's radio, the pigeon that lived on the sill and had opinions.

The pencil was 4H everywhere. Hard, grey, thin. It does not smudge and it does not forgive.

Book Five, 2076, a fortnight of levelling along the Kimberley reef line. Gogo had noted, on the third day, *staff-man late, drunk, not again* and then on the fourth day the same man's initials and no comment, and then on the ninth day, *P. Motaung's back — sent him home, did staff myself, closure 4 mm over 1.9 km, acceptable*. Naledi had never heard of P. Motaung. He had a back and a bad week in 2076 and here he was.

There were tram tickets used as bookmarks. There was a receipt for a tyre. In Book Seven a child's drawing of a horse had been folded flat and left at the page for station 44, and the horse had five legs, and on the back in ballpoint someone had written *NALEDI 4* in a hand that was not Gogo's.

She looked at that for a while and then put it back where it was and turned the page.

Book Two was 2071.

She had expected to find the landing in it. There was nothing about the landing. There was station 118 on the eleventh and station 119 on the fifteenth and between them four days of no stations, and a note on the left page, in the temperature column, that said *hot*.

That was very Gogo. Her mother had told a story about it — the whole country on the roofs and Dineo Mokoena on the reef with a level, and afterwards, when the family asked her, she had said the admin took longer than the ship.

Book Two, station 131, was the north-west quadrant. Bearings, chain

distances, the vent shaft, the two survey pegs, the collar of the old Number Six ventilation adit with its year of sinking. Everything closed. Everything in the right column.

The sketch on the right page ran a long spur south-east from the collar, dog-legging twice, the way an adit does when it is following something and not going somewhere.

The spur stopped mid-page.

Not at a margin, not at a fold, not at the end of a line. It stopped in white paper with a good two inches of clean sheet below it and no station number, no coordinates, no *ends*, no *fallen*, no *not entered*, no fudge factor in capitals. The 4H line came down and stopped and there was a dot where the pencil had been lifted, slightly heavier than the line, the way a hard pencil leaves a dot when the hand stays a moment after it has decided.

Naledi turned to the back of the book where the station index was, in Gogo's handwriting, all the way through, all nine books, every number accounted for.

131. 132.133.

She turned back. She counted the pages. She checked the front and rear pastedowns for a knife line, because the honest thing to check first is whether a page has been cut out, and no page had been cut out; the signatures were sound and the sewing was original.

She held it up to the window and turned it, in case there was an impression from a sheet written on top of it. There was the ghost of station 132 coming through from the other side, and nothing else.

She looked at the spur again. Sixty metres of it, drawn to scale, bearing 138 degrees magnetic, with the two dog-legs sketched at the angles they actually were, because Gogo did not straighten things to make a drawing tidy.

Then white paper.

Her grandmother had put a man's name in a book for overcharging on a chain. She had recorded a staff-man's back.

Naledi sat with the book open on her knees and her tea gone cold on the desk and did not know what she was looking at, and knew exactly

how much of it there was.

“Ma. When she stopped surveying. Was it her eyes?”

“It was 2081 and she was sixty-four and the Council put everything on the machines.” Her mother was washing a pot she had already washed. “She said the machines were fine. She said she wasn’t going to hold a staff for a boy with a tablet.”

“Did she keep going down?”

“Down where?”

“Underground. After.”

Her mother stopped with the pot in her hand. “She went to the funerals. Miners’ funerals. Every one she heard about, until she couldn’t walk to the taxi.” She set the pot on the rack. “I asked her once why she went to strangers’ funerals and she said they weren’t strangers.”

“That’s not an answer.”

“It was hers.”

She took Book Two and left the other eight in the chest, and her mother watched her do it and said, “Bring it back.”

“I will.”

“Not in a year. Bring it back.”

At the gate the pigeon came off the sill and went over the roof and the radio next door was doing the Renewal, some panel, a man saying the word *sovereign* twice in a sentence.

On the taxi Naledi put the book flat on her knees and opened it to station 131 again and photographed the page with her phone, badly, twice, and then a third time square.

Then she sat with her thumb on the white part of the sheet, which told her nothing, because paper does not tell you anything. She wrote in her own notebook: *NW quad, adit 6 collar, spur bearing 138M, 60m + ?*

She looked at the question mark. Then she wrote the bearing again, in the margin, so it would be on two pages.

The keeping house, Langlaagte • five weeks to Renewal

The Stewards do not have an office. They have a house, and you cannot phone it.

What you do is put a request in the custody box at Gate Nine, on paper, and a Steward reads it, and if the answer is yes somebody tells you at your work. Naledi wrote hers on a stall service log because she had nothing else in her bag: *Tuning house, N. Mokoena. Question about the north-west quadrant and about the 2064 keeping roster. I can come any day after four.* She folded it and put it in the slot and felt like a child posting a letter to a farm.

Four days. Then Karabo, at the tea urn, with the air of a man handing over contraband: “Somebody left this at the desk for you. It says Thursday.”

Langlaagte on a Thursday afternoon smells of dust and cut grass and the burnt-sugar coming off the reclamation plant. The fence runs the whole way along Main Reef, three metres of it, mesh with the port’s green stripe woven through, and along the fence there are people, because there are always people.

The pilgrims had the shade side. Nine or ten of them, a folding table, a paper icon of a face everybody knows, printed cheap and creased at the eyes. A woman was pressing coins into palms as you passed. Naledi took one because refusing costs more time than accepting.

Brass, thumb-sized, warm from the woman's hand. G on one face, in the profile the century has settled on, which nobody who knew him thought looked like him. On the other face, two words, stamped shallow: *Be kind*.

"What would he have signed, sister?" the woman said. Not aggressive. She had a stack of the icons and a bucket for the printing costs and a child asleep on a blanket under the table. "That's all we're asking. Volume doubled, cycle shortened. What would he have signed?"

"I don't know."

"He'd have asked who was in the room." She said it comfortably, the way you say a thing you have said four hundred times and still believe. "That's all. That's the whole devotion."

Forty metres on, on the sun side, the other lot had a trestle and a solar printer and a paper spike, and a man in a Codehouse T-shirt was reading numbers off a sheet at conversational volume, over and over, to nobody in particular.

"Uplift forty-four C. Nine items. Ledger receipt ends four-seven-two-D. Payment schedule entry, page eleven of the technique annex, published, dated, hashed. If a man tells you the tonnage is secret, ask him for the hash. Just the hash. He won't have it."

He caught her eye and held up the sheet, hopeful.

"I know that one," Naledi said.

"Then you know more than the panel on the radio." He was young and sunburnt and pleased. "No offence to the coin people. Their heart is fine. But you can't audit a face."

The pilgrim woman, up the fence, was singing something now with two others, low, in Sesotho, and it was not a hymn Naledi knew, and it was good.

The keeping house is a mine manager's house from 1934 with the veranda glassed in. There is a lemon tree, a Toyota under a tarpaulin, and a brass plate by the door with nothing on it but a number.

Ma-Rethabile Khumalo answered with her sleeves pushed up. Eighty-one, short, hands like a woman who had done something with them

for sixty years, which she had: forty of it on the winders at Number Four.

“Mokoena,” she said. “The tuning house. You wrote on a service log.”

“I didn’t have paper.”

“Good.” She stood back. “Come. I’m doing the candles and I’m not stopping.”

The room off the passage had a long table, and on the table were candle stubs in a bread crate, and a saucepan, and a stack of new wicks on card. Ma-Rethabile sat and went back to it: melt, pour, set, trim. The dip-hooks were on a wooden rack blackened at the tips. A young man was at the other end of the table with a ledger tablet and a face like a man waiting to be introduced.

“Tefo,” he said, and shook her hand across the crate. “I’ve read your form twelve.”

Naledi put her bag down. “It’s closed.”

“I know. That’s why I read it.”

“Sit,” said Ma-Rethabile. “Hold this.”

So Naledi sat and held the mould steady while the old woman poured, and the wax came up the tube and the smell was clean and animal, and she found she had been talking for four minutes about ninety-one guide segments before she had planned to say anything at all.

Ma-Rethabile poured a second. “Trim those three.”

Naledi trimmed the three.

“Your grandmother was Dineo.”

“Yes, mma.”

“I met her twice. She wouldn’t take tea.” The old woman set down the saucepan. “Bring the light.”

The room at the back had been a pantry. It had a slate floor, no window, and shelving up to the ceiling, and it was cold the way rooms are cold when they are on ground and not on foundations.

Ma-Rethabile did not explain the room. She took things off shelves and put them on the slate.

A drum-recorder chart, unrolled and weighted at the corners with two spanners and a stone. Twelve hours of it, pen-ink on waxed paper, the trace flat and thin for the first third and then, from a point marked in pencil, wide — a band, not a line, the pen thrown so hard it had torn through in three places. Along the bottom in a clerk's hand: date, 14 March 2064, and the hours, and *Coupling Gallery, No. 2 pen, drum speed 1 rev/hr.*

A roster. Foolscap, ruled by hand. Names down the left, four columns to the right: down, up, hours, *voice*. Forty-one names. Beside eleven of them, in the last column, a tick. Beside three, a dash. Beside one, nothing at all and a bracket drawn around the whole line in a different ink.

A pair of shoes, size six, women's, the soles worn through at the ball of the foot and the uppers stiff with something pale.

A tally book of the sort a shift boss uses, open at a page of five-bar gates.

A tin of Vaseline.

"That's the fourteenth," Ma-Rethabile said, of the chart, and nothing else. She moved the stone two inches so the paper lay flatter.

Naledi crouched. She could read a trace. She had been reading traces since she was nineteen. She could see the pen going wide and she could see, in the wide part, the periodicity — regular, coarse, about a two-hour beat — and she could see it damp out and go wide again, four times, and then, in the last hour of the drum, the band narrowing to a line as thin as the start and staying there.

"What was the drive," she said.

"I don't know what you mean by drive."

"What was making it move."

Ma-Rethabile picked up the roster and set it down closer to Naledi's hand. "Forty-one people went down. Eleven of them were singing. Three had been singing and stopped."

"That's not a mechanism."

“No.”

“Mma, I’m an engineer. Two hours of beat at that amplitude is energy from somewhere, and eleven people in a gallery is not—”

“I know what eleven people is.” The old woman’s voice did not go up. “I was one of them.”

Naledi shut her mouth.

“You asked about the roster. That is the roster.” She turned it a quarter-turn, squared it to the edge of the slate, and left it. “The rest is sealed and I did not seal it and I cannot open it. The Court holds one part and the Accord holds one part and I hold this part, which is the part that is objects, and objects I can put on a table.”

On the shelf above, behind a folded blanket, there was a wooden case the size of a bread tin with a wax seal on the hasp gone brown and a Court sigil pressed into it. Naledi looked at it for exactly as long as it took Ma-Rethabile to notice her looking.

“Yes,” the old woman said. “That one stays shut.”

Tefo had followed them in with the tablet under his arm.

“Ask her the other thing,” he said.

“Ask me what.”

“Whether she’ll put any of this on the ledger.” To Ma-Rethabile: “Mma, she came to us. She’s guild. She’s already been closed once. If we’re ever going to state a reason, this is the—”

“Not in the pantry, Tefo.”

“Then in the kitchen, but let me say it in front of her, because I’m tired of saying it to you alone.” He turned to Naledi and he was not performing; he had the flat delivery of a man on a loop. “Two consignments halted in twenty-seven years. No reasons given. Not to the Commission, not to the counterparty, not on the ledger. Do you know what that looks like from outside this house?”

“Mysticism,” Naledi said. “Or money.”

"Mysticism or money. Thank you. Half the port thinks we hear voices and the other half thinks somebody paid us, and the Firstborn have decided we are the chosen keepers of the sacred blood, which we have refused four times in writing and they read the refusal as modesty." He tapped the tablet against his leg. "A reason on the ledger can be checked. It can be argued with. It can be wrong and corrected. A silence can only be interpreted, and everyone interprets it in their own favour, and we don't get a vote. Mma, we are handing the movement a relic. Us. The order that was miners first."

Ma-Rethabile rolled the drum chart, slowly, from the far end, keeping the waxed paper off the torn places.

"In seventy-eight," she said, "we halted a consignment. Ninety kilograms, formed, ready. We gave no reason. Within a month there were eleven people who had been in the gallery that year whose names were being read on the radio, and one of them lost her house, and another was met in a car park in Benoni by two men who wanted to know what she had heard." She slid the roll into its sleeve. "If I had put my reasons on the ledger, my reasons would have contained the reason we halted, and the reason we halted was a thing three people said, and I would have had to name the three people or write a sentence nobody could check. You want a reason that can be argued with. So do they. That is what an argument is for — you can win it. Sometimes what I have is not strong enough to win and it is still true."

"So we keep it because it's weak."

"We keep it because it is not ours." She put the sleeve back on the shelf, level with the others. "You will get your ledger, Tefo. You will get it while I am still alive, probably, and I will sign it and I will not be sorry. But you must be able to say the reason in one sentence and mean every word, and at Renewal, right now, you cannot. Neither can I."

Tefo looked at the shelf for a while. "That's a better answer than the one you gave me in June."

"I've had since June."

At the passage door, going out, Naledi stopped.

There was a survey sketch pinned to the wall by the coat hooks, unframed, the paper gone the colour of weak tea. No station number, no coordinates, no title block. A collar, a shaft, a spur running off at a shallow angle with two dog-legs in it, and a scale bar drawn in by hand with the divisions stepped off by dividers.

4H pencil. Hard, thin, grey. Where the spur ended there was no annotation, and the line stopped in clean paper with a dot at its end, slightly heavier, where a hand had stayed a moment after it had decided.

“Where did this come from,” Naledi said.

“A box,” Ma-Rethabile said. “In the winter of '64. Left at the gate at Number Four with the tally books. Whoever brought it did not stay.”

“It’s not signed.”

“No.”

Naledi looked at it a while longer. The bearing was not written anywhere on the sheet. She had to hold her hand up and estimate it off the north arrow, and she made it about a hundred and forty degrees, and she knew perfectly well how badly you can lie to yourself with a hand held up against a wall.

She did not photograph it. She could not have said why.

“Come back when you know what you want,” Ma-Rethabile said at the door. “Not when you know what you think.”

On the fence walk back the pilgrims had grown to twenty and the printer man had gone home, leaving his paper spike on the trestle with the sheets still on it, weighted with a brick, for anyone.

Naledi turned her phone on at the taxi rank and it gave her fourteen days of nothing and then one thing.

ACCORD COMPLIANCE OFFICE — COURTESY NOTIFICATION. *Ref: Renewal secretariat audit trail, cycle 3. Your ledger read-queries for the period 12 July – 2 August have been aggregated into the pre-signature audit bundle in accordance with annex 9(4). Queries listed: 2,114. Categories: reconciliation records (uplift), forming-shop registry, custody index (sealed — metadata only), survey and cadastral*

(Council legacy). No action is required of you. Should you wish to record a purpose statement against these queries, one may be lodged at any time before signature.

Two thousand one hundred and fourteen. She had not known the number. It was, she thought, about right.

The bit that stayed with her on the taxi was not *no action is required*. It was the word *categories*. Somebody had sat with her queries and sorted them into four piles, and one of the piles was the survey and cadastral, which had nothing to do with tuning, and nobody had had to be told to look for it; the categories were just there, ready, in the form.

She read it again at Bree.

Then she lodged the purpose statement, because you may as well be the one who writes your own file. She typed it standing up, one thumb, in the queue for the rank.

Purpose: reconciling family-B guide segments (form 12, closed) against forming reference. Survey queries: personal, family papers, north-west quadrant. Happy to discuss.

She read that last part and left it in.

The road south, and the Atlantic • four weeks to Renewal

The consignment went out of Lancaster on eleven crates and one ledger seal, and the seal was the part that could not be replaced.

The crates were ash, dovetailed, lined with felt and pine shavings, built by a man in Bird-in-Hand who had built cider presses all his life and asked once what the wood was for and then did not ask again. Two hundred sixty-one kilograms of formed gold: guide segments, four collars, and the long items that the forming shop at Gap called horns and the manifest called nothing at all. Ruth had weighed every crate herself on the co-op's beam scale, because the beam scale did not need power and did not need trusting.

Her signature was on the manifest under the words RECEIVED AND CLOSED.

They went south by rail. Traction platcaps, two hundred and forty tonnes of train, charging four hours at every siding where the wind farms sat, forty kilometres an hour on the good stretches and twelve where the roadbed had been relaid over the old right-of-way by hand crews in the seventies. The bed had piezo bolts under the ties. Somebody had done the sums and the bolts paid for the signal lamps and nothing else, and everyone in the Corps knew that and everyone in the Corps still said the railway ran on its own bumping.

Beiler counted the crates at every stop. Eleven. He counted them at Perryville, at Ashland, at Rocky Mount. It is not that he doubted

the count. Counting is what you do with your hands when you cannot sleep.

The interior south of the Potomac is a country of two kinds of place.

There are the towns that came back. Brick main streets, mesh nodes on the church steeple, a market with awnings and a scale, draft teams at the rail, a diesel or two under tarpaulin for the county's emergencies. Barn raisings still going, seventy years on, because there are still barns to raise.

Then there are the substation yards.

They cut the copper out of those in the sixties and the steel in the seventies. What is left are the transformer hulks, too heavy to move and too full of oil to burn cleanly, sitting in fields of Johnson grass with their bushings gone and their tanks split where they cooked. The pylon lines were felled and taken for the steel, and what is left of them is the stubs, cut off knee-high, in rows, going over a rise and out of sight. Fifty kilometres of stubs, sometimes. Sumac between them.

Ruth watched them go by from the open door of the freight car with her boots hanging out and her cap in her lap. A private named Yoder sat down beside her and looked at the stubs.

"My grandfather says it took a minute and a half," he said.

"To burn?"

"To burn the whole east."

"It took about eleven hours," Ruth said. "The first minute and a half was the part he saw."

Yoder thought about that and then said what everybody says, which is that it seems a lot of iron to leave lying, and Ruth said the oil was the problem, and they left it there. Behind them Beiler was singing something with the words out of it.

She was six in March 2064. What she has of it is a set of small things in a fixed order. Her father coming in from the barn in the middle of the day, which he did not do. The kerosene lamp brought out of the cellar and the glass being wiped with a cloth, and the smell of the wick catching. Her mother taking the butter out of the propane cooler and

putting it in the springhouse, and the crock being too heavy for her mother, and Jonas carrying the other side.

Nothing burned in their district. That is the whole of the Plain memory of the Flare and it makes the rest of the country furious to this day. Nothing burned, because there was nothing there to burn.

The hunger years came from the outside, on foot, up the Newport Road.

They embarked at Charleston out of the deep berth the Corps keeps at North Charleston, on the *Ashley Fair*, sail-hybrid bulker, four wing-sails and a platcap bank amidships and wave-flex panels down her flanks that make her sound, in a beam sea, like a woman shifting in a chair.

Twenty-one days to Cape Town if the weather is ordinary.

The gold went into number three hold, forward corner, chocked and strapped, with the ledger seal photographed against a dated board and hashed and pushed onto the mesh before the hatch came over. After that the seal was the ship's problem and Ruth's problem and nobody else's on earth.

Watch on the hold was two hours. Ruth put herself in the rotation. Beiler told her a captain does not stand hold watch and she said she was a quartermaster, and he said that was not an answer, and she said no it wasn't, and took the midnight.

The log lives on a shelf by the ladder. Time, temperature, relative humidity, seal intact yes/no, initials.

Twenty-one days of that.

The temperature ran steady. The humidity went where the weather went. The gold did not care about either; the felt did, and the felt was hers.

Nights, the hold hums.

Not the gold. Hull noise coming down the frames, and the wave-flex skin working, and a bilge pump somewhere aft that cycles every

nineteen minutes and does not want fixing. Ruth sat on an upturned crate of nothing with the lamp turned to its low notch and her back against a stanchion, and after the first week she stopped bringing anything to read.

Her grandmother's Bible was in her kit and stayed there. The German in it is a language Ruth can hear better than she can see. On the ship she found she did not need the page. *Der HERR ist mein Hirte, mir wird nichts mangeln.* She would get four lines in and lose the fifth and go back to the start, and that was the watch, some nights, and it went quickly.

Other nights she had the coin out.

It is the one from the Ephrata pour, 2071, thick and badly struck, the rim gone soft from thirty years of thumbs. The Sunday version says *Be kind* and is what the children get. Hers has the whole of it, both sentences, the first one with the word in it that the county assemblies stopped reading aloud in about 2085 and that her grandmother, who was not Fold and never would be, said at least you know what they mean.

Ruth ran her thumb over the first sentence in the dark of number three hold, where nobody could see her do it, and put it back in her breast pocket.

The hunger years, as Ruth carries them, are not a famine. They are a queue.

The Compact granary for their district was at Ephrata and the district fed, at the end, something over eleven thousand people who had walked in from Reading and Philadelphia and one family from Trenton with a wheelbarrow. Her mother ran a soup line for four years and never once used the word charity, because it was a work rota and you were on it.

What Ruth remembers is the smell of wet wool drying in the assembly room and the sound of a hundred and forty people not talking while they ate.

In 1976 there was a hymn her mother liked. She has that wrong. It was not 1976.

Her brother's name is Jonas. He is two years older. When the Army moved their end of the district into the granary compound in the second winter, on account of the road people, somebody had to be left with the animals, and Jonas was fourteen and stayed with them, alone, for five weeks in a cold barn with a Springfield he was not permitted to load. That is not the staying that the family means when they say it, but it is where it started.

Ruth left at eighteen and enlisted, and Jonas stayed. He is on the home place at Gordonville with a wife and six and a hundred and forty acres and eight mules, and he has never once said anything to her about the Army, in twenty-three years, and that is not a coldness. He asks after her feet. His letters are one page, weather and stock and prices, and at the bottom, always, *we think of you*.

She writes back longer than he does. She has always written back longer than he does.

Day fourteen, the cook lost the ship's cat for a day and a half and had the whole after-guard turning out lockers, and it was in the sail locker having kittens, and the cook came up the ladder holding two of them like a man holding evidence. Beiler put it in the ledger of kindnesses. Yoder asked what the entry was and Beiler said, feeding a nursing mother, four hands, and Yoder said that was the cat's kindness not his, and Beiler said he was not going to argue theology with a private on a Tuesday.

Ruth laughed until she had to put her hand on the rail.

Day sixteen the wind went round and the wingsails came in and they ran four hours on the bank, and the hold got warm, and the humidity climbed nine points in a watch, and she wrote it down, and it came back.

That watch she thought about the bins at Christiana.

It was the wet fall of 2089. She was a first lieutenant and the district's movement officer, which meant she wrote the schedule that said which wagon went where and when the steel bins came empty for the next intake. That year the wheat came in at nineteen, twenty

per cent moisture, and the plan out of Corps was to bin it wet and run the forced-air driers, because the driers were new and the Corps had built them and the numbers said they would carry it.

The threshing men said stook it. Two of them, older, from Peach Bottom, said it in a room with her in it: leave it in the field on the stook, lose four per cent, keep the crop. They had a hundred and sixty years of doing that and no paper.

The plan stood. Ruth did not object to it. She scheduled it.

Bin four bridged. Wet grain does that; the crust arches over and hangs, and under the crust is a void, and the bin looks full and moves nothing. The order for the crust was poles from the top hatch, and the men on the crust were roped, and one of the ropes was tended by a man who had not been trained to tend a rope because Ruth's schedule had him tending a rope at 0400 and driving at 0600.

The crust went. Grain flows like water and weighs like grain.

Three men. Enos Zook, who was fifty-one. Christian Zook, his nephew. And a boy from Gap named Aaron Fisher who was nineteen years and four months and had been on the rota nine days.

Ruth was in the scale house forty metres away with a clipboard.

They pulled Christian out alive in the first ten minutes and he is alive now, in Kinzers, and he waves at her when she goes by.

The board of inquiry found the plan sound and the execution deficient, and named the deficiency at the level of the movement schedule, and her captain took the finding onto his own record, which was his right and which she had not asked for. She had asked for the opposite. She was reassigned to Fort Indiantown Gap and promoted eleven months later, on time.

The Fisher family came to her. That is the part. They came to the barracks in a spring wagon, the father and an aunt, and they brought bread, and they sat in the visitors' room at Indiantown Gap and told her she was welcome at the funeral and that they held nothing against her, and she said thank you, and they went home.

She did not go to the funeral. She has never been able to give an account of that.

Twenty years on she can say what she learned, and it is small and useful and does not weigh anything like the thing it came from: when the men with a hundred and sixty years and no paper are saying one thing and the driers are saying another, you stop the schedule and you take the four per cent.

Nobody has forgiven her, because nobody has ever accused her, and she has noticed that those are not the same and that she would prefer the accusation.

She wrote 0200, 24.1, 71%, seal intact, RS.

Day nineteen the wind came back and the sails went out and the ship leaned into it and Ruth stood at the rail before her watch with the coin in her fist and the Atlantic going by black.

She had a job at the far end of it. Deliver eleven crates. Get a receipt. And be able to tell the county assembly at Gordonville, when she stood up in a plank chair with her hat off, what the gold had been: what they paid, what they were offering, or what they were handing over. Three different sentences, and the manifest supported all three, and her people would want the true one.

That was what she had. Twenty-one days of humidity in a book and a question she could not answer with a scale.

She went down the ladder at four minutes to midnight and Yoder came up past her and said the humidity was falling and she said good, and he said, Ma'am, and she said, Get some sleep.

eGoli Spaceport, Gate Seven • three weeks to Renewal

The *Ashley Fair* came alongside at Cape Town on a Tuesday and the crates came up to Johannesburg on the Corps' own flatbeds under a Compact escort of four, and the whole thing was so ordinary that Ruth spent the rail leg waiting for the part where it stopped being ordinary.

Customs was a woman with a tablet and a cold. Gate Seven was a scanner arch, a weighbridge, and a queue of two.

The queue of two was because a Chilean consignment had come in the day before and somebody's forming certificate had the wrong annex number on it and now nine crates of Chilean gold sat in a bonded stall while two men argued about a footnote in the sun.

"How long," Beiler said.

"Since yesterday," said the gate clerk, cheerfully. "It'll be fine. It's page numbering."

Naledi got the liaison posting by list. It went up on the board outside the tuning house at half past six with the week's rotations, printed, because Kagiso believed in paper for anything a person might want to complain about.

Consignment US-11 (Lancaster/Gap). Berth 7C. Tuning liaison: N. Mokoena. Deputy: K. Sithole.

She read it twice. Karabo read it over her shoulder and made a noise.

“Two weeks ago you were closed,” he said.

“I know.”

“And now you’re liaison on the biggest inbound of the cycle.”

“I know.”

“So either nobody spoke to anybody, or somebody spoke to somebody and decided the safest place for you is inside the room with the door open.”

“Karabo.”

“I’m just saying you should know which it is.”

She did not know which it was. Kagiso, when she found him at the urn, said only that the Americans had asked for somebody who could read a family-B segment on sight and there were four people at the port who could and two of them were on the Chilean mess and one had a new baby. He said it while pouring, and he did not look up, and Naledi thought that was probably true and probably also not the whole shape of it.

Berth 7C, eleven o’clock. The port’s noise at that hour is the gantry crane and, under it, the vitrified bore breathing.

Nobody calls it breathing on paper. On paper it is the thermal cycle: the bore’s wall gives up heat to the shaft air on a period of about forty minutes and takes it back, and the air moves, and where the catwalks cross the throat you feel it on the back of your neck going one way and then, later, the other. Naledi had walked that catwalk twice a shift for six years. The wall is glass. Nobody has ever made glass like it and nobody has explained the wall, and it is a stanchion mount and a place people leave their coffee.

She got there before the truck and laid out.

Comparator on the bench, levelled. Two accelerometer pucks, cabled, calibration date checked against the sticker and then against the ledger because stickers lie. The reference rod in its case, family

B, 47 millimetres, kept at twenty degrees in a box that costs more than her flat. Thermal blanket. Log tablet. A jar of chalk for hands.

Then she waited eighteen minutes, which is the actual work of tuning liaison.

The truck came in with the Compact escort walking. The escort walked because a man walking beside a load can see a strap go slack and a man in a cab cannot; Naledi worked that out about four seconds after wondering why nobody had ridden.

The officer was mid-forties, plain-faced, hair cut off short under the cap, sleeves down and buttoned in thirty-one degrees. She had the manifest in a hard folder, actual paper in a plastic sleeve, and she was reading the crane's certificate plate on the way past like a woman who read plates.

"Captain Stoltzfus. Quartermaster, US consignment."

"Naledi Mokoena. Tuning liaison." They shook. The captain's hand was dry and had a callus across the top of the palm that Naledi could not place.

"You'll want the seal photographed before I break it."

"Please."

"And I'd like the weighbridge slip put against my beam-scale figures before we go further, if that's not a nuisance."

Naledi looked at the eleven crates. "You weighed them on a beam scale."

"Each one. In Lancaster."

"The weighbridge here is good to two hundred grams on twenty tonnes."

"So is a beam scale," Ruth said, "and mine hasn't got software."

There is a kind of person who says that to be difficult. Naledi watched her say it, flat, with no lift at the end, and decided she was not that kind. "Fine. Karabo, get the slip."

They broke the seal at 11:40. It came up on the ledger inside eight seconds — hash, timestamp, the tuning house's counter-signature — and Ruth stood and watched the screen for the whole eight seconds with her thumb on the manifest and then let out a breath through her nose and put the folder under her arm.

"That's mine done," she said.

"Not quite."

"No. But that's the part I could lose."

Crate four came out first because crate four had the horns in it, and Naledi wanted a horn on the bench before she touched a segment, for reasons she would have found hard to justify to Kagiso and did not have to.

The lid came off and the pine shavings came out by the handful into a bin, and there was the felt, and under the felt the long items lay in cradles cut to them.

They are 1.4 metres. They are not horns. The manifest calls them nothing at all: item, mass, purity, form reference. The forming shop at Gap calls them horns because they taper and because men name things. What they are is a tapered waveguide section with a flare at the wide end and a machined collar at the narrow, and the flare is not a curve you would arrive at by taste.

Naledi lifted one out on two hands with the chalk on her palms.

Nine and a bit kilos in the hands and it wants to keep going in the direction you started it. She set it in the cradle on the bench, adjusted, and put the puck on the collar with a smear of couplant.

"How long's the run," Ruth said.

"Twelve minutes for the thermal to sit. Then four minutes of it, then I do the same to the next one, and the next one, and there are nine, and then the collars, and then the segments, and we will both be here at eight tonight."

"Good," Ruth said, and sat down on a crate that was still full.

Here is what a calibration is, for anyone who has to sit through one.

You have a piece of gold formed to a shape. You strike it, softly, with a nylon-tipped hammer at a marked point — the same point, the same energy, the port has a jig for it — and you record what it does. What it does is ring at a fundamental and at a stack of overtones above that, and the ratios in that stack are the form's signature. Purity sets the losses. Geometry sets the ratios. If the forming shop has done its work the ratios sit inside a tolerance you can print on a card, and the card is what the counterparty's manifest specifies, and that is the entire trade: a shape that rings correctly.

You also get the noise. The bore breathes, the crane runs, a truck goes past on the perimeter road. That lives at the bottom of the trace and you learn to see through it the way you learn to see through a dirty windscreen.

Naledi struck the first horn at 11:56 and watched the comparator draw.

Fundamental clean. Second, third, fourth clean. Everything inside the card, which she had expected, because Gap's forming shop was reputable and because the beam-scale woman on the crate would not have brought her bad gold.

And at 2.31 times the fundamental, where the card says nothing at all should be, a line.

Not noise. Noise is a hedge. This was a picket: narrow, twelve decibels up out of the floor, sitting there for the whole four minutes of the run and not caring what she thought about it.

She looked at it for a while.

"Something's wrong," Ruth said.

"Why do you say that."

"Because you stopped moving."

Naledi struck the second horn.

Same picket. Same place. She did the third, and the fourth, and she stopped keeping her face arranged around the fifth because it was costing attention she wanted elsewhere.

Nine horns. Nine pickets at 2.31, within a fifth of a per cent of each other, in nine pieces of gold formed by hand in Pennsylvania by three different men on two different mandrels over eleven weeks.

Which is not possible from forming.

Forming error scatters. Nine pieces off the same mandrel will scatter across the tolerance like shot; nine pieces off two mandrels will scatter in two clusters. Nine pieces agreeing to a fifth of a per cent on a feature nobody was aiming at is not the shop's hand. It is something the shape *is*, or something the room *does*, and she had one way to tell which.

She swapped in the reference rod. Family B, 47 mil, twenty degrees, the port's own, formed in Germiston, never nearer Pennsylvania than this bench.

Struck it.

The picket came up at 2.31.

Naledi put the hammer down on the bench and left her hand flat next to it.

Ruth had been watching for two hours and had understood roughly a third of it, which was enough.

She understood that a line on a screen had appeared in a place where the woman had not been looking, and had then appeared in the same place nine more times, and had then appeared in a piece of metal that had nothing to do with Ruth's crates at all, and that this last had been worse rather than better.

She got up off the crate.

"Say the part I need."

Naledi did not look away from the comparator. "There's a feature in the response at two point three one times fundamental. It isn't in your gold and it isn't in mine. It's in the room."

"The room."

"Something in this facility is ringing at a frequency that shows up in everything I put on this bench, and it is not in the tolerance card, and it has been logged before, by me, three weeks ago, on the Chilean line." She turned her head. "That log was closed."

"Closed how."

“Reviewed and closed. Form twelve. It happens; most of what we file is nothing.”

“Is this nothing?”

Naledi said, “No,” and that was all she said, and Ruth noticed the woman’s tempo had gone from six words a second to two.

Ruth stood and looked at the trace for a while, because it was there and because she has always believed that if a thing is going to be difficult you should look at it early.

Then she asked the question she had, which was not any of the ones Naledi expected.

“Was that line in my crates when I loaded them?”

Naledi opened her mouth and shut it. “I don’t know.”

“Can you know?”

“Not from here.”

“Then here’s what I’ve got.” Ruth put the folder on the bench and opened it flat, and turned it around, and it was the beam-scale sheet: eleven crates, tare, gross, net, in pencil, in a column, with her initials against each line and the date. “Two hundred sixty-one kilograms and forty grams. Weighed twice, four days apart, in a stone barn at Gap, on a scale I cleaned myself. Formed on the ninth through the twenty-first of June. Three shops, and I can name the man for every item and the day he pulled it. The horns came off Ammon Stoltzfus’s mandrel and off Yoder’s, and Yoder’s is the older one and I know its number.”

Naledi looked at the column.

“Why are you giving me this?”

“Because you can tell me what the gold does and you can’t tell me where it’s been, and I can tell you where every gram of it has been to the hour.” Ruth tapped the sheet once with two fingers. “If the line came aboard at Charleston, that’s the ship. If it came at Cape Town, that’s the rail. If it was there in the barn in June, then it was there before your room existed for us, and we have got something else entirely. And there is a way to find out, which is that Gap struck every item on their own jig before crating, and they wrote it down,

because the annex makes them, and those sheets are in a drawer in Pennsylvania.”

There was a silence with the crane in it.

“You’d get those for me,” Naledi said.

“I’d get them for the manifest,” Ruth said. “The manifest is mine. You can read over my shoulder.”

They ran it again at four, with the crane stopped, which cost the port money and took Kagiso two phone calls and a favour.

The bore’s noise floor at 2.31 with the crane stopped is nineteen decibels lower than with the crane running. The picket did not move. It sat there twelve up, as clean as a rung bell, in an empty berth with nothing in it but two women, a bench, and eleven open crates.

Naledi wrote the run number in the log by hand, and then the ledger reference, and then a sentence, and then she read the sentence over and did not soften it.

Feature at 2.31 F reproduced on nine consignment items and on port reference rod, crane stopped, noise floor -19 dB. Not attributable to forming. Not attributable to instrument. Origin unknown.

She put her initials. Then she turned the tablet around.

“You’re not required to sign that.”

“No.”

“You’d be signing that you watched it.”

“That’s the only part I could sign.”

Ruth read it twice. Then she took the stylus and wrote, under it, in the witness field, *Present for runs 4 through 13. R. Stoltzfus, Capt., USA, quartermaster, consignment US-11*, and the time, and handed the stylus back.

Neither of them said anything about it. There was nothing to say that would not have been smaller than the thing.

Ruth went back to the Corps' rooms at Gate Seven at nine and sat on the bed and got her boots off and did not lie down.

She had a signature on a manifest that said RECEIVED AND CLOSED, and she had a receipt coming, and the plain reading of her orders was that in eleven days she would hand over eleven crates, take the paper, and go home and stand up in a plank chair at Gordonville and say what she was told the gold had been.

There was a version of the next three weeks where she did not mention the line.

It was not even dishonest. Nobody had asked her. A quartermaster's business is mass, seal, condition, and delivery, and the tuning was the host's affair, and a woman who has spent twenty-three years in a logistics corps knows exactly how much of the world runs on people not volunteering.

She got the coin out and turned it over once and did not look at it, because you do not need to look at a thing you have known with your thumb for thirty years.

The first sentence is about this. It has always been about this. It is about the specific and available choice to be the sort of person who lets a room close over a true thing because the paperwork does not require otherwise.

She put the coin back in her breast pocket and buttoned it and got her writing case out, and wrote to Gap for the jig sheets, and to Beiler with the escort roster for the week, and then, because she had the case open, one page to Jonas. Weather, and the ship's cat, and that the country here was drier than she expected and the mules were bigger.

She did not write anything about the line. She would not have known how to say it in one page.

Naledi stayed until the maintenance crew came to lock the berth and then walked out along the perimeter and did not take the taxi.

She was thinking about the reference rod, which was the part that had gone through her like cold water. Nine horns from Pennsylvania she could argue about all week. The rod was hers. The rod was

Germiston steel-town gold, formed six years ago, kept in a box she had personally logged the temperature of four hundred times, and the rod had the same picket in it, which meant the picket had been in every calibration she had ever run at this port and she had been looking straight through it for six years, the way you look through a dirty windscreen.

Twelve decibels up. Not hidden. Not subtle. Just outside the card, and everything outside the card is nothing, because that is what a card is for.

Her hands were still chalky. She wiped them on her trousers, which is against the standing instruction, and she stopped at the fence where the reclamation plant makes the air smell burnt and got her tablet out.

She did not file. Filing had been done and closed once and she had learned something from that, and the thing she had learned was not *don't file*.

She opened the survey directory instead, the personal one, and she typed the bearing she had estimated off a wall in Langlaagte with her hand held up like an amateur, and the two dog-legs, and the length of the spur as she had stepped it off the hand-drawn scale bar, and she saved it, and she gave the file a name.

2.31.

Then she went and got the taxi, because it was nine forty and she was not going to walk to Bree.

eGoli Spaceport, the tuning house • three weeks to Renewal

The Chilean footnote got fixed at eight in the morning and the whole port learned about it by nine, because the fix was that somebody in Santiago re-issued the certificate with Annex 4(b) instead of 4(a) and a nine-crate consignment sat in a bonded stall for thirty-one hours over one letter.

Karabo found this hilarious for about a day and a half.

“Thirty-one hours,” he said, “for a bracket.”

“For the wrong bracket.”

“Same bracket. Different letter.” He was filling in the humidity column for the stall log with a pen because the stall log is one of the four things at eGoli still kept on paper. “You know what the Chileans are going to do now.”

“Re-issue everything with both.”

“They’re going to put both annexes on every certificate for a year and then somebody will write a note saying you can’t cite two annexes and we’ll do it all again.”

Naledi laughed, and then she stopped laughing, because at half past nine the rotations went up on the board and her name was on Chilean re-certification.

It was not a demotion. That was the first thing she checked and the thing she kept checking for the rest of the week, the way you keep touching a tooth.

Chilean consignment CL-09, re-certification support: N. Mokoena (lead), K. Sithole.

Consignment US-11, tuning liaison: K. Sithole (acting), deputy T. Mahlangu.

Lead. She had been deputy on nothing and now she was lead on something, and re-certification support is real work — nine items, full family-C run, the annex trail rebuilt from the forming shop's sheets up. It is exactly the work she had asked for on US-11. It was two berths away from US-11 and it would take four days and she would spend those four days with her hands on Chilean gold, which had also had the picket in it three weeks ago, and which she was now being paid to certify.

Kagiso was at the urn again. He is always at the urn; she has begun to think the urn is where the tuning house does its actual business.

"You gave the American line to Karabo."

"Karabo's ready."

"Karabo's ready and I'm on Chilean."

"You're lead on Chilean." He said it the way you hand somebody a coat.

"Kagiso."

He put the cup down. He is fifty-eight and has the guild's whole hand-history in his fingers, three knuckles that don't close and a thumb that got between a collar and a cradle in 2081. "Two things are true," he said. "One. The Chilean thing needs somebody who can read family C and argue about annexes, and there are two of those in this building and one of them is me and I'm in Renewal briefings until the fourteenth. Two. Commissioner's office rang the scheduler on Monday to ask whether the liaison postings were fixed or indicative."

"That's not a no to me."

"They didn't ask about you. They asked whether the postings were fixed." He picked up the cup again. "The scheduler said indicative."

Which is true. The scheduler is not a liar.”

“Kagiso—”

“I know what you’re going to say and I’d like you to say it somewhere that isn’t the urn.”

The Commissioner’s office is in the administration block on the east side, three storeys, brown brick, built 2078 and looking it, with a lobby full of the guild’s old comparators in glass cases and a mural of the first berthing that nobody likes.

Dr Sefako Marais came out to get her himself. That was the first thing he did.

He is sixty-one, heavy through the shoulders, and he was in shirt-sleeves with his cuffs turned back twice, and there was a stack of ledger printouts on the corner of his desk fifteen centimetres deep with a paper knife lying on top of it. He moved a chair for her. He asked whether she wanted rooibos or the other thing, and when she said the other thing he made a face and said good, because the rooibos was ceremonial and eight years old.

“Mokoena. Dineo’s granddaughter.”

“Yes.”

“She surveyed the Rand Deep Levels re-entry when I was a graduate. I sat in the back of a hearing once while she took a mining lawyer apart over a bearing.” He sat. “She did it entirely in numbers. He didn’t understand he’d lost for about a week.”

“That sounds like her.”

“How is she?”

“She died in ninety-eight.”

“Ah.” He put his hand flat on the desk for a second. “I’m sorry. I did know that. I was at the Codehouse funeral for Mabaso the same month and I have the two of them filed together and wrongly.” He looked at her. “That’s not an excuse, it’s just what happened.”

She had come in with a sentence ready and now the sentence was in the wrong shape.

"You logged a feature at two point three one," he said.

"Yes."

"On the Chilean line three weeks ago and again on the American items last Thursday, and on a reference rod, with the crane stopped." He had not looked at a page to say it. "Reproduced fourteen times. Not attributable to forming. Origin unknown. An American captain witnessed it in the field and initialled, which was her right and slightly unusual."

"You've read it."

"I read everything with 'origin unknown' in it. There are about forty a year." He turned his hand over. "Nine of them are somebody's phone. Four are the reclamation plant. One year we had a beautiful one at eleven hertz that turned out to be a fan in a canteen freezer that had been running out of balance since 2091. Do you know how long that took to find?"

"No."

"Fourteen months. And the man who found it is now a section head and deserves to be." He leaned back. "So. What do you want?"

She had rehearsed this too, and this part still fit. "Instrument time. Twelve hours on the gallery array with the port at Sunday load, so I can get a bearing on it. And the sealed sixty-four telemetry, because if it's the bore or the node I need to know what a rough wake looks like and there's no other record of one."

"Right." He said it without a flinch, which frightened her more than a flinch. "The instrument time is a scheduling question and I don't own the schedule; Renewal owns the schedule until the sixth of next month, and Renewal has nine delegations wanting gallery walk-throughs and a Chinese survey party who have been promised the throat catwalk twice and denied it twice. You'd be applying for a window against those. You should apply. The form is a fourteen, and it goes through the guild-master, not me, and you should ask Kagiso to sign it rather than filing it up the chain, because a fourteen that arrives without his signature comes back down to get it and that's ten days."

She wrote 14 on her tablet.

"The sixty-four telemetry is not mine either. It's Steward-sealed, and the Accord's confidentiality annex sits on top of the seal, and the release path is a joint application — guild and Stewards, both, and then Sela's office is notified, not consulted, notified, and the notification has a thirty-day clock on it." He said it evenly. "Thirty days from the twelfth is after Renewal."

"Then waive the clock."

"I can't waive the clock. I can ask the Elder Steward to shorten it and she has refused four such requests since I've held this desk, including one from me, and she has never explained a refusal to anybody, which is either the most irritating institutional habit on this continent or the reason we still have a threshold chamber. Pick one on the day of the week."

Naledi sat with that.

"You're not saying no."

"Why would I say no?" He looked genuinely puzzled, and she believed the puzzlement. "It's an unexplained twelve-decibel feature in the medium the entire species relationship runs on. If it's real I want it on the ledger with a mechanism next to it. If it's a fan in a canteen freezer I want that on the ledger too, with the fan's serial number." He pulled a printout off the top of the stack, glanced at it, put it back. "What I'm saying is that I have twenty-one days, and in twenty-one days I have to get a volume schedule, a technique schedule, a settlement mechanism and a custody clause past nine delegations and a Chinese bloc that has spent three years arguing that Africa is holding the species' door for rent. And two of those delegations are looking for a reason to say the host is unfit."

"This isn't a reason to say that."

"It is exactly the shape of one." He said it kindly. "A discrepancy in the host's own calibration chain, logged, closed, re-logged. I could write the Chinese note myself. Three paragraphs. It would be a good note."

"So you want me to sit on it for three weeks."

"No. I want you to be right about it for three weeks and then say it in a room where it can't be used." He put his palm on the printout

stack. “The difference between those two things is enormous and it is going to feel to you, for three weeks, like no difference at all. I know that. I’ve been on your side of it. I lost that argument in 2081 and I was right and the man who beat me was wrong and I’ve since come to think he was right about the timing, which is a horrible thing to have to know.”

She looked at the stack. She could read the top sheet upside down: it was a volume table, tonnes per annum by year, out to 2130, with somebody’s pencil in the margins.

“We can be curious for twenty-seven years,” Marais said. “We can be sovereign for six weeks.”

“That’s a good line.”

“It’s a true line. Those aren’t exclusive.” He stood, which ended it. “File the fourteen. Get Kagiso to sign it. And Mokoena — if it comes back with a window in it, take the window at Sunday load and don’t tell me about it in advance, because if you tell me in advance I have to tell Sela’s office, and I’d rather find out afterwards like everybody else.”

At the door he said, “The bearing you want. Off what wall?”

“Langlaagte. An old stope wall.”

“Hm,” he said. “Is that hers?”

“It’s her handwriting.”

“Then it’ll be right,” he said. “She was never wrong about a bearing. She was wrong about people.”

She filed the fourteen on the Tuesday afternoon. Kagiso signed it in front of her, and read it first, all of it, and asked her to change one word — *bore* to *facility structure*, because a fourteen naming the bore goes to the geotechnical committee, which meets monthly and had met.

It came back on Thursday.

Not rejected. Returned for completion: item 9(c), *estimated effect on scheduled Renewal activity*, left blank, and item 11, *co-signature of*

berth authority for affected berths, unsigned. She had left 9(c) blank because she did not know the answer and had written *nil expected*, and *nil expected* is not a figure. The berth authority is a man called Pieterse who is not obstructive and was in Durban until Monday.

She re-filed on Monday with 9(c) filled in as *twelve gallery-hours, Sunday load, no berth movements affected* and Pieterse's signature under it.

It came back on Wednesday. The Sunday she had asked for was gone. A Chinese survey party had the throat catwalk from 06:00 and gallery isolation could not be guaranteed with a party in the throat, which was true, and Naledi checked it, and it was true.

The scheduler offered her the following Sunday.

The following Sunday was five days before Renewal.

Nobody had said no. She wrote it out on the back of a canteen napkin that evening, all of it, the way you write out a fault tree: office asks scheduler whether postings are indicative; scheduler answers honestly; guild-master allocates by competence; Commissioner encourages the application; form returns twice for genuine omissions; berth authority genuinely in Durban; Chinese party genuinely promised the throat twice already. Every link load-bearing. Every man in the chain doing his job well.

The tree came out clean and the answer at the bottom of it was *after signature*.

She put the napkin in her pocket and did not throw it away, and later that week she found it gone soft in the wash.

The tuning house has a technical meeting on Wednesdays at four. Twenty-two people, an urn, a screen, and Kagiso running it off a half sheet.

She had the item put on: *anomalous common-mode feature, cross-consignment, request for coordinated logging protocol*. Which was small. It was the smallest thing she could ask for: that every calibration in the house, for the next three weeks, record the amplitude at 2.31 F in a named column, so that at the end of three weeks there would be four hundred data points instead of fourteen.

It cost nothing. It cost one column in a log.

She presented it in six minutes. She had the fourteen traces up and the reference rod trace last, and she said the sentence about the card and the windscreen, which she had not meant to use in a meeting and which landed.

Then the room.

Thandi Mahlangu said the column was fine but the naming was a problem: if you name a column *2.31 anomaly* in a house log, the house log is discoverable under the Accord's audit annex, and in eighteen days there will be nine delegations' technical staff reading every discoverable document at this port, and they will find a column headed *anomaly* in four hundred calibrations and no explanation next to it, and the Commissioner will spend a day of Renewal on it that he does not have. Call it a spectral note. Log the number, not the word.

Which was reasonable. Naledi said fine, call it whatever you like.

Then Sipho Nkosi, who is a good tuner and has been on the choir line nineteen years, said he did not want to log a column he could not defend. If he writes down an amplitude at a frequency, he is asserting a measurement, and the pucks are not calibrated for narrowband work at that frequency, and if the number is soft and somebody builds an argument on four hundred soft numbers, that is on him. He would log it if the house bought narrowband channels. The house has not budgeted narrowband channels.

Which was also reasonable, and Naledi, who had been irritated for four seconds, stopped being irritated, because she had thought the same thing about the pucks at eleven o'clock the previous night and had decided the ratio was robust even if the amplitude was soft, and she had not written that down anywhere, and if she had not written it down then Sipho was right to refuse.

Then somebody asked what the mechanism was, and she said she didn't know, and there was that pause that a technical room does when it has stopped being with you.

Kagiso said: no protocol change three weeks out from Renewal; log it as a free-text note at the tuner's discretion; revisit at the first meeting in the new cycle. He said it in nine seconds. He was not unkind about

it. He wrote *revisit* on his half sheet, and she saw him write it, and she believed he would.

The item took eleven minutes of a two-hour meeting. The next item was the canteen's coffee contract and it took twenty-five.

Sipho caught her at the door with his cup still in his hand.

"You're right about the ratio," he said.

"You just said—"

"I said I won't log an amplitude. The ratio's clean. A fifth of a per cent across fourteen pieces off four mandrels. You can't get that from forming and you can't get that from mounting." He drank some of the coffee, made a face, and put the cup on the sill. "I've seen it. I've seen it for years."

Naledi stood very still. "In runs?"

"In runs. It's above the card so I've never written it down. Neither has anyone. There's nowhere to write it." He shrugged, one shoulder. "I've got two on the choir line and a daughter doing her third year at the academy, and if I stand up in a meeting eighteen days out from Renewal and say the port's calibration chain has an unexplained common mode, the sentence leaves the room without me attached to it and comes back in a Chinese technical note. Then Kagiso has to defend me, and he will, and it costs him. And in ten days he has to go into a room and get the guild's custody language into the amendment, which is the only thing at this port I actually care about."

"So you'll say nothing."

"I'll say it to you at a door." He picked the cup back up. "That's not nothing. It's just not a meeting." He looked at her, straight. "Bring me a mechanism and I'll say it in the meeting with my name first on it. I'm not brave, Naledi. I'm arithmetic. Bring me a mechanism and the arithmetic changes."

He went off down the corridor. Two others said versions of it to her that week, in a car park and over the urn: Thandi, who told her to keep every trace off the house server and on her own hardware, and gave her a spare drive out of her bag; and a young tuner called Lebo who

said nothing useful at all and left a printed 2081 committee minute on her bench with a paragraph highlighted, and never mentioned it.

Nobody in that building was a coward. That was the part she could not get comfortable against. They all had a thing they were carrying and were carrying it well, and the aggregate of twenty-two people carrying their things well was a door closing without a hand on it.

Friday she went to look at the Chilean gold, because it was hers to certify and because she had begun, in a way she did not like, to think of berths as places to be near.

Nine items, family C, formed at Rancagua. She ran the full set and the picket was there at 2.31 in all nine, exactly where the American horns had it, and she wrote it in free text at the tuner's discretion, as instructed, in the box provided, at the bottom of a certificate that was otherwise entirely clean.

Then she signed the certificate, because the gold was good. That was the true thing about it. The gold was good, the annex was now cited correctly in both brackets, and the Chileans could ship.

Karabo came past at six with two rolls and put one on the bench.

"How's the American liaison," she said.

"Fine. She writes everything down twice. She weighed the crates again after unloading."

"Of course she did."

"She asked me what the two point three one column on my log was for." He tore the roll in half without much interest. "I said ask Naledi. Was that wrong?"

"No," Naledi said. "That was exactly right."

She ate half the roll and looked at the certificate on the screen with her name at the bottom of it and the true sentence sitting in a free-text box where nobody would ever query it, and she thought about the napkin, gone to pulp in her trouser pocket, and every good man in the chain.

Then she got out her tablet and opened the file called 2.31 and added a line: *Sunday 26th. Five days out. Take it.*

74EGOLI SPACEPORT, THE TUNING HOUSE · THREE WEEKS TO RENEWAL

eGoli Spaceport, Berth 7C and the Gate Seven mess • three weeks to Renewal

The jig sheets came in from Gap on the Monday mesh drop, forty-one pages, scanned crooked, and Ruth printed them on the Corps' machine because the Corps' machine was there and because paper can be laid side by side.

She laid them on the mess table at Gate Seven. It seats twelve. There were three of them at it: Ruth at the head with the sheets, Beiler at the far end doing the escort roster because he had nowhere better, and Naledi standing over the middle with her thumbs in her belt loops looking at Amish penmanship for the first time in her life.

"They wrote the strike energy," Naledi said.

"They wrote everything. The annex makes them."

"No, the annex makes them write the fundamental. Nobody writes the strike energy. I don't write the strike energy, we have a jig, the jig is the strike energy." She bent closer. "This is a hammer drop from a marked height. He's written the height. He's written the height in inches."

"Nineteen and a quarter."

"Why does he know that in inches."

"Because the barn is in inches," Ruth said. "Everything at Gap is in inches and everything on the manifest is in metric, and I converted

every line twice on two different days and I'd rather you check me than not."

Naledi straightened up and looked at her for a second and then said, "You converted two hundred and sixty-one kilograms of gold by hand."

"By hand and then on the tablet, and where they disagreed I did it a third way."

"Did they ever disagree?"

"Twice. Both times the tablet."

Beiler did not look up from the roster. "It's always the tablet," he said. "She tells everyone that. It's been the tablet twice in twenty-three years."

"Amos."

"Ma'am."

Here is what they had, laid on the table, and it is worth being exact because the exactness is the whole thing.

Naledi had the port's instruments: comparator, pucks, the jig, the reference rod, and six years of runs on a house server she was now keeping copies of on Thandi's spare drive. What her instruments told her was *what the metal is doing now*, to a fifth of a per cent, at a resolution nobody in Pennsylvania could touch. What they could not tell her was anything at all about time. A calibration is a photograph. She had a beautiful photograph of a thing and no idea when it had walked into the frame.

Ruth had the provenance: three shops, eleven weeks, mandrel numbers, the man's name against the day, beam-scale masses in pencil, the rail waybill, the ship's hold plan with the crates' stow positions marked, the Charleston bond stamp, the Cape Town customs entry, the flatbed run sheet. What her paper told her was *where every gram had been to the hour*. What it could not tell her was whether any of it rang.

Neither toolkit was interesting. Together they were a clock.

"Right," Naledi said. "Fundamentals first. Read me item one."

"Item one. Segment, family B, four seven, mass eight-one-oh-point-four grams, fundamental" — Ruth turned the sheet — "he's written it as a note frequency and a number under it. Two thousand nine hundred and six point one hertz."

"Note frequency."

"It's a shop that also makes bells."

Naledi got her own trace up. "I have two nine oh six point four."

"Three-tenths."

"Three-tenths over eleven weeks, a ship and two rail legs is nothing. That's temperature and my couplant." She wrote it down. "Next."

They did all thirty-one items in an hour and ten. Every fundamental agreed inside half a hertz. Every second, third and fourth overtone agreed inside the card. The gold that had left the stone barn was, in every respect the annex cared about, the gold now sitting in eleven crates in Berth 7C.

Then Naledi said, "Now the part I actually want. Did he log above the fourth?"

"He logged everything he saw."

"Ruth. Nobody logs everything they see."

"Turn to the back of each sheet."

There was a back of each sheet. On the back of each sheet, in the same hand, in a column with no heading, were three or four more numbers, and against several of them a small mark like a tick that had thought better of itself.

Naledi put both hands flat on the table.

"What's the mark?"

"I don't know. I've asked. He'll write back in about eleven days."

"Read me the column for item one."

"Six eight one nine point oh. Then eight seven one eight, that one's got the mark. Then nine nine—"

"Stop." Naledi did not look up. "Say the second one again."

“Eight seven one eight.”

She did it on the tablet where Ruth could watch, which was deliberate; she had learned in four days that the captain did not trust a number she had not seen arrive. Eight seven one eight over two nine oh six point one. She turned the screen around.

2.9998.

“That’s three,” Ruth said.

“That’s the third overtone, which is in the card, which is fine, which I don’t care about.” Naledi wiped it. “Do the fourth.”

Nine nine nine four over two nine oh six point one came out at 3.4390 and Naledi said a word in Sesotho that Beiler, who had spent nine days on the rail leg with a Sotho brakeman, recognised and did not comment on.

“It isn’t there.”

“What isn’t.”

“Two point three one. Ammon Stoltzfus struck every one of these items on his own jig in a barn in Pennsylvania in June and wrote down four numbers off the back of each one and not one of them is at two point three one times fundamental.” She sat down on the bench, hard. “It wasn’t there in the barn.”

Ruth thought about it for a moment.

“Or he couldn’t see it,” she said.

Naledi opened her mouth.

“You told me it’s twelve decibels up off the floor,” Ruth said. “In an empty berth with the crane stopped. What’s a barn’s floor? He’s got a stone barn with a road outside it and a forge running twenty feet off and a scale I cleaned myself, and he’s reading a fundamental off a bell-tuner’s ear and a meter that’s older than me. If your line is twelve up off *your* floor, is it twelve up off his?”

Naledi sat there.

“No,” she said. “It’s twelve up off mine because mine is nineteen down with the crane stopped.” She put her palm over her eyes for a second. “His floor’s probably thirty up on mine. It could be sitting

in there at full amplitude and he'd never separate it. God. I've just spent an hour proving a negative I can't prove."

"You've proved the fundamentals travelled."

"Which nobody doubted."

"I doubted," Ruth said. "It's my manifest."

They ate in the mess at two because that was when the mess had food. Mutton, and the port's bread, and a dish of the beans Beiler had been trading a Compact ration tin at a time to the Malawian crane crew for.

"So we can't date it from the barn," Ruth said. "Then date it from the ship."

"You didn't strike anything on the ship."

"No. But I've got the stow plan, and I've got a hold that flexes, and I've got three weeks of a wave-skin generator log because the engineer wanted my signature on the fuel draw." She got the folder. "And I've got where every crate sat, and which ones were against the frame, and which two were in the middle of the stack with gold on all sides of them."

Naledi stopped chewing.

"Say that again."

"Crates six and seven were centre-stack. Gold above, below, and both sides for three weeks."

"And the others weren't."

"Two and eleven were against the frame. The rest were edge or corner."

"If it's something the gold does to gold — if it's coupling — then six and seven should be worse." Naledi put the fork down. "Not different. Worse. Amplitude, not ratio. And if it's the room here then all eleven crates are identical because they've all been in the same room for four days."

"Is that a test?"

"That is a test." She was up. "That's a real test and it takes three hours and I can run it on Karabo's berth time because he's on annex work until Thursday and he owes me for the Chilean bracket."

Ruth got her cap. "You'll want the stow plan."

"I'll want the stow plan and I'll want you to read the crate numbers off the tags, out loud, because I will absolutely convince myself I'm looking at seven when I'm looking at six." Naledi stopped in the doorway. "Also I need to say — the reason I keep asking you for paper is not that I think you've made an error."

"I know that."

"I want to be clear, because on Wednesday I asked you three times whether the seal was intact and you went quiet."

Ruth held the door. "I went quiet because I was doing the hold plan in my head," she said. "You've never once suggested I lost the load."

"Good."

She hadn't gone quiet about the seal. She had gone quiet because on Wednesday, for eleven or twelve seconds, she had considered the possibility that a man at Gap had put something in a crate, and had had to walk the whole roster of them, three shops, every name, before she could set it down. She was not going to say that in a corridor. Naledi walked out of the mess believing the misunderstanding was closed, and Ruth believed she had answered the question, and neither of them noticed that they had just had two different conversations.

They ran it from four to seven, crate by crate, and Ruth called the tags.

"Six."

"Six." Naledi struck. Waited. "Twelve point one up."

"Seven."

"Twelve point three."

"Two. Against the frame, port side."

"Twelve point one."

"Eleven."

"Twelve point two."

It came out flat across all eleven crates inside a quarter of a decibel, which is the same as flat. Three weeks with gold on every side of it had done nothing to crates six and seven that four days in Berth 7C had not done to everything.

Naledi sat back on her heels and looked at the comparator and laughed, once, without much in it.

"It's the room," she said. "It was always the room. I've now proved it twice with two different sets of somebody else's homework."

"Is that bad?"

"It's the second-worst answer. The worst answer would have been that it's the gold, because then I could point at a crate." She rubbed chalk off her thumb. "The room means it's the facility, and the facility is the bore, or the node, or something under the node, and the only instrument that reaches down there is Steward-sealed and I've got a Sunday five days before signature."

Ruth put the stow plan back in the folder and squared it.

"Then I've been useful in the wrong direction," she said.

"You've been useful. You closed the ship. I'd have wondered about the ship for a fortnight and there'd have been a version of me at the conference getting asked about it by a Chinese technical officer who'd read the hold plan." Naledi looked up. "That's not nothing. That's a door you shut."

"That's provisioning," Ruth said. "You shut doors behind a column so nothing follows it."

"Is that a saying?"

"It's a manual."

At half past seven the berth lights went to night level and the bore did the thing it does, and the air on the catwalk went from moving

one way to moving the other, and Ruth felt it on her neck and turned her head like a woman hearing a step behind her.

"That's thermal," Naledi said. "Forty-minute cycle. It's been doing it since 2071 and nobody has explained the wall."

"Nobody has explained the wall."

"It's glass. Nobody makes glass like it. It's a stanchion mount and a place people leave coffee." Naledi packed the pucks in their foam. "You get used to it."

"I've been getting used to things since I was six," Ruth said, and did not expand, and Naledi did not ask, and that was the correct handling and both of them knew it.

They sat on the empty crate for a while because the tablet was still writing out the run.

"Your grandmother," Ruth said. "The one whose handwriting you keep looking at."

"Dineo."

"Was she a surveyor like a man walking a fence, or a surveyor like an office?"

"Like a fence. Instruments and boots. Forty years in the deep levels." Naledi turned the tablet over in her hands. "She left me eleven survey books and there's a spur drawn in the margin of one of them that isn't on any map at this port, and she never once mentioned it, and she had sixty-two years to mention it."

"Did you ask?"

"Twice. She changed the subject twice and I was fourteen and it was a Sunday and there was food."

Ruth nodded slowly. "My brother stayed," she said. "Sixty-nine. Everybody else came out to the Compact and he stayed on the home place with the Ordnung as it was, and he's alive and he's in Lancaster and I've written him one hundred and eleven letters and he's answered nine."

"Do you know why nine?"

"No," Ruth said.

Neither of them said anything else about it. The tablet finished writing.

Then Ruth went into her breast pocket and put a coin on the crate lid between them.

It was not a gesture with a run-up. She put it down the way you put down a tool you have finished with, and she was already reaching for her folder.

Naledi picked it up. It was heavy for its size, brass-coloured, worn to a shine on both faces from a thumb, with a raised edge and a face on it she recognised from about four hundred pieces of tat sold outside the west gate to pilgrims — G, in profile, done badly.

“They’ve made him look like a bank manager.”

“They’ve made him look like whoever’s cutting the die that year,” Ruth said. “Turn it.”

The reverse had a spine of small linked squares around the rim, which she knew was supposed to be the ledger and looked like a bicycle chain, and inside it a line of text in two rows.

The gate tat had one row. *BE KIND*. She had read it every day for six years and never read it.

This one had both rows.

She held it in the night lighting and read it twice and her thumb went, without asking her, to the worn place where thirty years of somebody else’s thumb had been.

“The gate ones don’t have the first part.”

“No,” Ruth said. “The gate ones are for tourists and the ones in the soft churches at home are worse, they’ve got a face bigger than the sentence.” She took the folder under her arm. “It’s two sentences and the first one is the one that costs. You can’t do the second one on a Tuesday if you’ve spent Monday doing the opposite. Every place that drops the first line drops it because the first line is about *them*.”

“Is that a Fold thing to say or a Ruth thing to say?”

“It’s a thing I say and I get looked at for it.” She held her hand out, flat, palm up.

Naledi put the coin in it.

“Do the ones your people carry all have both?”

“The honest ones do.” Ruth put it back in her breast pocket and buttoned the pocket. “The pocket’s how you tell. If it’s on a wall, somebody’s decided which half of it you get to keep.”

She wrote it up that night at the little desk in the Corps’ room with her boots off.

Held crate-by-crate response test with tuning liaison Mokoena, Berth 7C, 16:00–19:00, on the crate stow hypothesis. Result: negative. No amplitude variation attributable to hold position. Ship’s carriage is not the source. Recommend the ship be excluded from further enquiry and that this be stated in writing before any delegation asks.

Then she thought for a minute and wrote a second paragraph.

Note for my own record. The engineer is a good engineer and the shops are good shops and my paper is clean, and none of that was ever the question. The question is what the load was walking into. I have been provisioning columns for twenty-three years and I have never before been asked to escort something to a place where nobody could tell me what it was for. I have not been lied to. I have been answered.

She looked at that for a while and did not cross it out.

Then she wrote to Ammon Stoltzfus at Gap and asked him what the small mark on the back of the sheets meant, and asked him to be exact, and asked him to write it in inches if inches was how he had it.

eGoli Spaceport, the Corps' room at Gate Seven • two weeks to Renewal

The order came in on the Tuesday drop with the mail, which was correct, because a recall is administration and administration travels with the mail.

It was four lines. *Consignment US-11 delivered and receipted. Escort detachment released. Capt. R. Stoltzfus and detachment to route Cape Town-Charleston, first available Corps hull, not later than 29 August. Report Lancaster on arrival.* Then the signature block, Colonel Deborah Reyes, Logistics Corps, Eastern Command, and a routing note saying berths had been requested.

Ruth read it twice, put it face up on the desk, and squared it with the edge of the desk, which is what she does with paper she is going to think about.

Then she read the rest of the drop, because you read the rest of the drop.

There were three other things and all three were about the same order, which is how the Army says a thing it means.

The first was from Reyes herself, personal, six lines, and no rank in it after the first word.

Ruth. The order's the order and you'll see it's not a rebuke. Volume schedule is signed off at our end whatever happens at yours, so there

is nothing left for a quartermaster at that port. You have been at sea or on a road for eleven weeks and I have your leave balance in front of me and it is obscene. Also: nine delegations. Nobody needs a US officer standing near a controversy in a host country. That's not me protecting the flag. It's me not spending your career on somebody else's argument. Come home. Take the leave. Then come see me about Fort Indiantown Gap, because Ammon's shop is going to be a bottleneck by 2103 and I want your assessment before someone in Washington writes one.

The second was the movement order proper, with the hull named. USNS Cordwood, sail-hybrid, Cape Town on the twenty-fifth.

The third was from the Compact office at Lancaster, and it was signed by a man called Ezra Zook, who is not Army and not clergy and holds a job that has no good English name: he is the person who writes to families.

Sister Ruth. The bishop's people asked me to write since I write better than they do. You will have the Corps' order by now. The word here is that the consignment was carried well and the barns are proud, which they will not say to your face. Your mother is fair. Your brother's Sarah was buried the second week of August, she was eighty-one and it was her heart, and Levi stood at the grave and looked well enough. He did not ask after you. I am telling you because you would rather be told.

One more thing and then I am done. There is talk that you may be asked to stand at that conference. I do not know if it is true. If it is true I would ask you to think whether standing there is your work or somebody's want. We have a great deal of practice at not being famous. It has served us.

Ruth put the three of them side by side on the desk with the order at the top and looked at the arrangement for a while.

Sarah Stoltzfus had taught her to card wool when she was five and had a laugh like a door. Eighty-one and her heart. Ruth wrote the date in the back of the Bible, in the column where the dates go, under her father and above nothing yet, and then she sat with her hand on the page.

Beiler came in at nine with the escort roster, saw the order, and read it standing up because she turned it toward him.

"The twenty-fifth," he said.

"Cape Town on the twenty-fifth."

"That's clean, then." He set the roster down. "Rail's running Thursdays. If we go the Thursday we've got two days at the dock and the men can be paid out at Charleston instead of here, which is better for them, the rate's better." He was already doing it, out loud, because that is what he does with news. "Sithole owes us a flatbed and I'd want the crates' empties back, they're Corps property and somebody at this port will absolutely lose eleven crates."

"Amos."

"Ma'am."

"Sit down a minute."

He sat, cap in his hands.

"If I stayed," she said. "Not the detachment. Me."

He did not look surprised, which told her the mess had got there ahead of the mail, which it always does.

"On what," he said.

"That's the question."

"No, ma'am, that's the answer." He turned the cap over. "You've got no orders to stay. You've got no berth here, no ration line, no billet after the twenty-fifth, and no mandate. You'd be a woman in a uniform standing in a room where nine countries are arguing, and every one of them would want to know who sent you, and the true answer would be nobody, and nobody is the worst answer you can give in a room like that. They'd fill it in for you."

"I know."

"You'd be a captain who refused a movement order. That's not a conscience thing on the paper, that's a paragraph and a board." He said it flat, not unkindly. "And you'd be doing it two weeks before your colonel wanted you in front of her about Gap, which is a job you'd be good at and which would put a shop in every county in that state

inside ten years. You want to talk about what the load's for. That's what it's for."

"You've thought about this."

"I've thought about it since Friday, ma'am, because I saw the engineer's face at the crate test and I know yours." He put the cap on his knee. "May I say the rest of it."

"Say it."

"I was at Nazareth in sixty-eight, the second winter, in a barn with three hundred people in it who'd walked out of Newark. What got them fed was that a colonel two hundred miles away decided which way the trucks went and everybody did what he said the day he said it. Not the day after. That's the whole thing. That's the only reason my sisters are alive." He shrugged, one shoulder, the same shrug he uses about weather. "Obedience isn't a small virtue. Everybody talks about it like it's the thing you do when you can't think. I've seen what it fed."

Ruth said nothing for a moment.

"Ammon's mark on the back of the sheets," she said. "You know what I think it is?"

"No."

"I think it's the number he wasn't sure of." She turned the order a few degrees on the desk and turned it back. "He wrote it down anyway and put a mark on it. He didn't leave it off and he didn't pretend it was clean."

"That's a barn, ma'am. Not a treaty."

"No," she said. "It's the same paper."

Sunday assembly at eGoli is in a prefabricated hall on the west side that the port put up in 2081 for the Filipino crane crews and then everybody else came, and it is now shared out across the day by rota: the Catholics at seven, the Fold at nine, the Zionists at eleven with drums, the Muslims across the way in a proper mosque built by the Malaysians in 2086 because they were the first to be organised about it.

The Fold hour at eGoli has maybe seventy people in it. Three Americans, and the rest port: Kenyans, a Brazilian family, two Chinese engineers who come and do not sing, a Malawian crane driver called Faith who reads better than anyone. There is no minister. Somebody reads a folded passage, somebody reads from the ledger of kindnesses, and then it is quiet for a long time, and then people say things.

Faith read the passage. It was the one out of the Sermon, folded — the ladder had come out of it three lines that stood, and they stand as three lines, and Ruth has heard them in four countries.

Then the ledger. A crane crew had covered a man's shifts for nine days while his wife was ill; a woman in the Cape had paid a stranger's port fee; somebody had sat all night with a Chilean welder in the clinic. Names, dates, what was done. Nobody thanked. You read it and you put it down.

Then quiet.

Ruth sat with her hands on her knees and her thumb on the coin through the cloth of her pocket and did not take it out.

After a while a woman she did not know said that her son had been rude to her and she had been rude back and she had thought about it all week, and sat down. After a while one of the Chinese engineers said, in careful English, that he was not of this and he came for the quiet and he hoped that was permitted, and Faith said it was, and he sat down.

Then Faith said, without getting up: "Captain. You've been sitting in that seat with a face on you for an hour."

Seventy people looked at her. Ruth stood up, because you stand.

"I've been recalled," she said. "The order's good. My colonel is right on every count and she is fond of me, which makes it worse." She put her hands behind her back, which is Army, and heard herself do it. "There's a discrepancy in the port's measurements that I have witnessed and initialled. My name is on it. I don't understand it. It may be nothing. If I go home on the twenty-fifth, I go home before anybody knows."

Nobody said anything for a moment.

"Then you'd be going home not knowing," Faith said. "That's most of us, most days."

"Yes."

"So what's the thing you'd be doing wrong."

"I don't know yet," Ruth said. "That's the honest answer. I can make an argument that staying is pride. I can make it easily. I'm a captain in a logistics corps, I moved a load and the load arrived and it was receipted, and the rest of it is nine countries' business and I have no standing in it, and if I stay I am standing somewhere in a uniform for a reason I've chosen myself." She stopped. "The other side of it is only that I saw a thing."

The Brazilian father said, "You could write it down and send it."

"I have written it down. Twice."

"Then it's done," he said, and he was not being unkind, he was being practical, and half the room's shoulders agreed with him. "Paper stays after the man goes home. That's the point of paper."

"That's the point of paper," Ruth said. "Yes."

An older man near the door, an American, one of the three, said: "Sister, I'd say be careful about the word *witness*. It's a good word and it's got a bad habit. It starts out meaning *I saw it* and after a while it means *I am the one who saw it*." He rubbed his knee. "I've watched it happen to men. In the soft churches at home they've got faces on the wall now bigger than the sentence. Every one of those faces started out as somebody who genuinely saw something."

"I know that," Ruth said.

"I know you know. I'm saying it in the room because it's true and it deserves saying."

Then it was quiet again, and the quiet went on longer than was comfortable, and nobody rescued it, which is how the Fold hour works and why she keeps going.

Nobody told her to stay. Not one person in that hall told her to stay. She had not expected them to and she was surprised anyway.

Naledi caught her at four on the walkway over 7C with a printout in her hand and an expression like a woman who has been walking fast.

“Pieterse signed off my Sunday, the twenty-sixth, twelve hours, and I’ve got Kagiso to put his name on the isolation request instead of mine which makes it a house job instead of my hobby—” She stopped. “You’ve got a face.”

“I’m recalled. Ship on the twenty-fifth.”

Naledi stood there with the printout.

“The twenty-fifth,” she said.

“The day before.”

“That’s—” She looked at the paper and then at the crates below and then somewhere else entirely. “Right. Okay. That’s not a coincidence in a way that I need to be careful about, because it looks like one and it almost certainly is one. It’s a hull sailing on a day.”

“It’s a hull sailing on a day,” Ruth said. “Reyes wrote it herself. She wants me home on leave and then in a room about Ammon’s shop, and both of those are real, and if you saw her letter you’d like her.”

“Are you going?”

“I haven’t decided.”

Naledi opened her mouth, and Ruth watched her decide not to say the thing, and respected her for it.

“You were going to ask me to stay.”

“I was going to ask you to stay and then I thought about what I’d be asking you to pay with,” Naledi said. “So no. But I’m going to tell you what I’ve got, because you initialled it, and then it’s yours.”

“Do that.”

They went down and she did it, standing at the comparator with the pucks still in their foam, twelve minutes, and it was clean and small and she did not oversell any of it. Sunday load. Twelve hours on the gallery array. A bearing off a wall at Langlaagte from her grandmother’s book, and if the bearing was right the feature had a direction, and if it had a direction it was not the room in general, it was a place in the room.

"And if it's a place," Ruth said.

"Then it's a place under the port that isn't on the port's map." Naledi put the last puck in. "And I go and ask a Steward about it and she doesn't tell me, and that's the twenty-eighth, and Renewal signs on the thirty-first."

Ruth looked at the eleven crates. Two hundred and sixty-one kilograms, formed in three shops, weighed four times, and now sitting in a bonded stall in a room that did something nobody could name.

"Thank you for not asking," she said.

"Ke sona," Naledi said. "Doesn't mean I'm not asking. It means I'm not saying it."

She wrote the day up at the little desk with her boots off, because she writes the day up.

Recall received, Corps hull 25 Aug, Cape Town. Detachment released. Beiler to prepare movement; rail Thursday. Empties to be recovered, eleven, Corps property.

Then she sat with the pen.

Sarah buried 2nd week Aug. Eighty-one. Her heart. Levi stood at the grave.

Then, further down, in the same hand:

Every argument for going home is a true argument. I have not been given a bad order or a stupid one. Nobody has asked me to lie. Reyes is right that I have no mandate. Amos is right that men eat because somebody moves the day he is told. Zook is right that we have practice at not being famous and it has served us. The old man at assembly is right about the word witness, and I have watched it happen to a man in Kandahar Barracks and it was ugly.

Against all that I have one line, which is that I was there and I initialled it.

I do not know if that is the first sentence or my own want wearing it.

She read it back. She did not cross it out and she did not add a conclusion, because there wasn't one, and putting one down would

have been the lie.

She turned the lamp off at ten. The coin was on the desk where she had left it, face up, both rows, and she did not put it in her pocket.

The Stewards' house, Langlaagte Road • two weeks to Renewal

The Stewards keep a house on Langlaagte Road that used to be a mine hospital. Two storeys, brick, iron roof, a jacaranda in the yard old enough to have been planted by somebody who thought the shaft below would run forever. Naledi came at seven in the morning on a Wednesday because Tefo had said seven and because she had a berth check at eleven and had allowed herself four hours for something that should have taken twenty minutes.

The gate was open. A woman was hosing the step.

“I’m here for Tefo Molefe.”

“He’s in the back with Ma-Rethabile,” the woman said, “and they’ve been at it since half five, so sit in the passage and don’t be shy about the tea.”

The passage had a bench and a bad print of the Vredefort ring and a table with a flask on it. She sat. She had the survey book in her satchel wrapped in a cloth, which she recognised, sitting there, as ridiculous—it was a book, it had lived in a garage in Meadowlands for twenty years—and she did not unwrap it.

The door at the end of the passage was not shut. It was pushed to, in the way of a door in an old building where the frame has moved and the door will not sit, and it had a gap of about four fingers, and voices came through it at conversational volume because the people speaking had forgotten the door.

She could have coughed.

“—then say it on Friday,” Tefo was saying. “Say it at the technical session. Twenty minutes. You don’t have to give them the logs, you give them the reasons. Sixty-eight and eighty-two. Why we stopped the consignment, what we were looking at, what we did next. That’s it. That’s the whole ask.”

“And when they ask why we did not say it in sixty-eight.”

“Then we say we were wrong not to.”

A chair moved. Ma-Rethabile Khumalo’s voice was low and unhurried and it did the thing that some old voices do where it does not rise to be heard, it simply stops leaving room for anything else.

“We were not wrong,” she said. “We were slow. Those are different animals and you keep putting the same saddle on them.”

“Ma.”

“Sit down, Tefo, you’re a man in a room, not a wave at the fence.” Papers. “You want stated reasons. Give me the sentence you want me to stand up and read on Friday. In front of nine delegations, the Commissioner, and four hundred press seats. Say the sentence.”

Silence in the room. Naledi found she had put her tea down on the bench beside her without drinking it.

“In August 2068,” Tefo said, “the Stewards’ array recorded a rise in the idle floor of the node coincident with the loading of consignment four. We could not exclude the possibility that the consignment was coupling to the node’s own resonance rather than only carrying it. We halted. Over the following fourteen months the floor returned. We released. We did not publish because we could not distinguish the effect from our own instrumentation.”

“Good sentence.”

“Then say it.”

“I will say it and by nightfall it will be two sentences,” she said. “The first will be *the Stewards halted a consignment because they feared the gold was doing something to the ground*. The second will be *the*

Stewards have known for thirty-two years and said nothing. And then there is no third sentence, Tefo, because nobody will wait for it. Your sentence has an *we could not distinguish* in the middle of it and that word does not survive a corridor.”

“So we let them have the corridor.”

“We have let them have the corridor for thirty-two years and the trade has run and nobody has been hurt by it.”

“That we know.”

“That anyone knows. Which is the same size of the world as your *we could not distinguish*, and you don’t like it when I use it, but it’s the same instrument.”

Naledi looked at the print of the Vredefort ring. Somebody had drawn concentric pencil rings on the mount, faintly, in the wrong place, and had not rubbed them out.

She did the arithmetic she had been avoiding since she saw the file come back closed. Thirty-two years of a rise in an idle floor. She had it in her own runs going back six. Her comparator did not reach the node; the Stewards’ array did. If they had a thirty-two-year series and she had six, then she was not the person who had found something. She was the fourth or fifth person, and the others had all had this same conversation in this same passage, and had lost it, or won it and then found that winning it changed nothing.

She got out her notebook and wrote *ask for the 2068 series, floor only, no interpretation* and put it away.

“Then here is what I actually want,” Tefo said, and his voice had changed, gone quieter and worse. “Halt it.”

A long pause.

“The Renewal.”

“Not the Renewal. The signature. Ask for a ninety-day deferral on custodial grounds. We have the standing—clause eleven, we’ve had

it since seventy-three and never once used it. Ninety days. The Commissioner can't overrule us without going to the Accord council and the council doesn't sit until March."

"And in the ninety days?"

"In the ninety days we publish. Everything. Sixty-eight, eighty-two, the floor series, the sealed sixty-four logs, all of it. Let the world's instruments look at our numbers. If we're wrong, nine countries' physicists tell us in a fortnight and we've cost the world three months. If we're right, we've stopped a species from signing a hundred-year schedule blind."

"Say the rest of it."

"There is no rest of it."

"Tefo. Say the rest of it."

Naledi heard him breathe.

"And the order stops being ceremonial," he said.

"There it is."

"That's not why."

"No," Ma-Rethabile said. "It's not why. It's what you'd get, and you'd take it, and you would be a better Elder for having taken it, and every single one of those things can be true while the halt is still a lie." Something set down on wood, a cup. "Do you know why I halted in eighty-two?"

"Because the floor moved."

"Because the floor moved and I did not know what I was looking at and I had an instrument and no theory, and the only honest thing available to me was to stop the load and keep watching. Fourteen months of watching. Nobody clapped. There was no press seat. Nineteen men in the deep levels ran that array on a two-shift rotation and eleven of them are dead now and none of them was ever interviewed by anybody about anything." Her voice did not lift. "What you are asking me to do is the same shape and none of the same substance. You are not asking me to stop and watch. You are asking me to stop *in front of people*. You want the ninety days because the ninety days

will be seen. And the moment a halt is for seeing, it does not matter at all whether the floor moved.”

“That’s unfair.”

“It is completely unfair and it is also what will happen.” A chair. “I have watched the other one for thirty years, so let me be even. Signing on the thirty-first because everybody is tired and the volume schedule is agreed and it would be rude to be the country that makes the room sit down again—that is a lie also. That’s the one your Commissioner will tell, and he will tell it kindly, and he’ll believe it, and it costs nothing today and everything in eighty years. I am not defending him. I am telling you there are two ways to be dishonest here and you have found the second one and you think it’s the cure.”

“Then what,” Tefo said. “Ma. What. Because you’ve just said we can’t sign and we can’t stop, and you’ve been saying it since June, and the calendar doesn’t care.”

“I know what the calendar does.”

“So what’s the third thing?”

“I don’t have it.”

“Then—”

“I don’t have it,” she said again, and it was flat, and it was the first thing in twenty minutes that had sounded like an old woman. “I have had eleven weeks and I don’t have it. If I had it I would have used it in eighty-two.”

Nobody said anything. The hose stopped outside. A taxi went up Langlaagte Road with its window down and somebody’s radio on it, a bass line and then gone.

Naledi sat very still on the bench, because in the last three days she had had a bad half hour in bed at two in the morning where she had thought the words *someone should stop it*, and had felt, thinking them, a warmth that she recognised now, in a passage, with a cold cup of tea beside her, as the wrong kind of warmth. She had been picturing a room. In the picture she was standing up in it.

She wrote in the notebook: *halt for seeing = signing for tiredness. Same paper.* Then, underneath: *no third option on the table. Not from them. Not from me.*

"The child is in the passage," Ma-Rethabile said.

Naledi stood up so fast she caught the tea with her elbow and did not knock it over, which she counted as a mercy.

"Ma-Rethabile, I—the door was—"

"The door has been broken since 2079 and I am the person who has not fixed it, so it's my fault and not yours." The old woman was in the doorway. She was smaller than the voice had been. Cardigan, boots, a lanyard with three keys on it and one of them a mine key from a shaft that had been backfilled before Naledi was born. "How much."

"From say it on Friday."

"So all of it."

"Yes."

"Hm." She looked at Naledi for a moment with something that was not unfriendly and was in no way soft. "Well. You've saved me a version of this where I have to be careful with you. Come in. Bring the tea, it's cold, Grace makes it at half four."

The room at the back had a table in it and eight chairs and a map wall, and the map wall was the reason nobody was ever brought into this room. Naledi did not look at it for longer than a second and looked at it for that second: workings in blue, workings in a red that had been laid over the blue by a later hand, and dates in the eighties in the margins in three different inks.

Tefo was standing with his hands on the back of a chair. He looked forty and tired and twenty-eight.

"You're the tuning liaison," he said.

"Naledi Mokoena."

"You filed a discrepancy on the twelfth of July and it came back closed on the sixteenth."

"You've seen my file."

"I've seen every file that gets closed in four days," Tefo said. "It's the only thing I'm actually good for."

She unwrapped the survey book on the table because there was no gracious way to do it and put it down open at the page, and put her finger next to the spur in the margin, and did not say anything about her grandmother, because she had rehearsed four sentences about her grandmother in the car and all four of them had been for effect.

"Dineo Mokoena," she said. "Deep-level surveyor, 2036 to 2076. This is her hand. The spur isn't on any port map I have access to. I have a bearing off the Langlaagte wall that puts my feature in that direction from the gallery array, and I have twelve hours of house time on the twenty-sixth, and I am here because I need to know whether that spur is a real place before I spend the twelve hours."

Ma-Rethabile did not touch the book. She looked at it from where she stood, hands at her sides, for perhaps eight seconds.

"It's a real place," she said.

Tefo turned his head.

"It's on that wall," Naledi said.

"It is on that wall."

"Then—"

"You have asked me a question I can answer, so I have answered it," Ma-Rethabile said. "Now ask me the next one and watch what happens."

Naledi asked it. "What's in it?"

"No."

There it was. Not a door slammed. A door that had been shut for a long time and had a chair against it and a person sitting in the chair, and the person was not angry, and would sit there all day.

"That's the whole thing," Tefo said, to nobody. "That's the whole order, right there. She'll tell you it's real. She won't tell you what it is. And the answer to why not is that she was told not to by a man who is dead."

"By four people," Ma-Rethabile said, "three of whom are dead, and one of whom is ninety-one and in Klerksdorp and will not see you either, and she was on the array in sixty-four with a lamp on her head and no idea whether the roof was coming down, and you will not use her in an argument in my house."

Tefo shut his mouth.

"Sixty-four," Naledi said.

"You want that too."

"I want to know if the floor I'm reading is the same floor they were reading."

Ma-Rethabile pulled a chair out and sat in it, which took her a moment, and Naledi understood suddenly that she was very old and that the voice had been costing her something for an hour.

"Nineteen of us went down in the worst week of the Flare," she said. "That's the number people say. It's wrong, it was twenty-three, four of them were pump men who came because the pumps were theirs. The trains had stopped. The pumps were on the mesh so the pumps were alive, which is the only reason any of this happened at all, and there is a whole religion in America about that and I don't have an opinion on it."

"Ma," Tefo said, "you don't have to—"

"I'm not doing testimony. I'm doing arithmetic." She put her hands on the table. "We were down eleven days. We came up. Something had changed and something had not, and I am not going to tell this young woman what I think it was, because the moment I tell her, she will have heard it from *Ma-Rethabile who was there*, and it will sit in her chest like a fact and it isn't one. Four of the twenty-three thought it was nothing. One of them thought it was everything and spent his last years being photographed at the west gate, God rest him, and they sell his face on a coin now with the wrong sentence on it."

She looked up.

"The logs are sealed," she said, "because when we came up we could not agree on what we had done. Twenty-three people in the same room with the same instruments. Do you understand what that means, as a technical fact, not as a mystery?"

"It means the observation was underdetermined," Naledi said.

"It means we didn't know." Ma-Rethabile nodded once, slowly. "And it means the honest record of that week is a set of instrument traces and twenty-three disagreements, and the world has spent thirty-six years wanting it to be a story. So it stays shut. Not to keep it holy. To keep it from being finished by somebody who wasn't there."

Naledi went out through the yard at twenty past nine with the survey book back in its cloth and Tefo walking her to the gate.

"She's right about me," he said, at the gate, looking at the jacaranda. "The ninety days. I've written the speech. I've written it four times and it gets better every time, and that should have told me something."

"She's right about the other one too."

"I know she's right about the other one. The other one's going to happen." He put his hands in his pockets. "Nine delegations, a volume schedule everybody's already banked, and a Commissioner who genuinely thinks he's carrying the world on his back, and he sort of is. On the thirty-first they'll sign it. Nobody will lie. Everyone will be tired and reasonable."

"So there has to be a third way to do it."

"There has to be," Tefo said. "Nobody in that house has one. I've asked her every week since June." He opened the gate. "Have you got one?"

"No."

"Right." He almost sounded relieved, which she found she disliked. "Come back on the twenty-eighth and I'll get you the floor series, sixty-eight forward, numbers only, no annotation. She'll allow that. She'll allow anything that's a number." He held the gate. "And Mokoena. If you find your grandmother's place on the twenty-sixth

and it's got something in it, don't come to us first. Come to us second."

"Who's first?"

"Somebody who can't be told to sit down," he said, and shut the gate.

She drove back to the port with the window open and got to the berth at ten past eleven, twenty minutes early, and stood at the head of the walkway over 7C for a while without going down.

Signing because everyone was tired. Stopping because stopping would be seen. Both of them were dishonest and both of them were on the table and there was nothing else on it.

She went down and did the eleven o'clock check, which was fine, all of it, every crate.

A garden above Hillcrest, KwaZulu-Natal • thirteen days to Renewal

The road up from Hillcrest is tarred for the first six kilometres and then it is not, and the hire car's suspension had opinions about the rest. Naledi parked on the grass verge behind a bakkie with a dead battery and walked up the drive between two rows of granadilla on wire.

The house was small. Brick, tin, a rainwater tank on a concrete pad, platcap bank in a louvred box on the north wall with a service sticker three years out of date and a bird's nest behind the box. Somebody had swept the step recently and not well.

A woman in her sixties came out and looked at Naledi's shoes.

"Mokoena."

"Yes."

"She's in the back. She'll give you an hour and she'll be finished at forty minutes, so don't take the first ten of it being polite." The woman held the door. "There's water. Don't let her stand up for the beans."

Priya Ellis was sitting on a plastic chair in the middle of a vegetable bed on a piece of old carpet, weeding, with a knife.

She was very small. The Assist frame showed only as a grey line

along the outside of one trouser leg and a collar at the back of her neck under the hat. Her hands were bad and worked anyway. She did not get up.

"You're the tuning one," she said.

"Naledi Mokoena. Guild of Tuning, eGoli. I wrote to you in July."

"You wrote to me in July and in August and the second letter was better." Priya put a weed on a pile. "Sit on the wall, the other chair's got a spider."

Naledi sat on the wall. Below the garden the valley went down in sugarcane and then bush and then a long way off there was the white line of the N3 and a truck on it, silent at that distance.

"I have your paper," Priya said. "The one your comparator wrote. Somebody sends me things. I read the numbers and skipped your discussion section."

"Why?"

"Because the numbers are yours and the discussion is the guild's." She cut something at the root. "The floor rises. You've got six years of it. The mine had thirty-two and wouldn't give it to you. Is that where you are?"

"Twelve hours of house time on the twenty-sixth, and a bearing, and a place under the port that isn't on the port's map."

"Whose bearing?"

"My grandmother's."

Priya's hands stopped, briefly. "Surveyor?"

"Deep-level. Thirty-six to seventy-six."

"Then she's the one to trust and not me." She went back to the weeds. "Ask your question. I'm eighty percent deaf on the left, sit further along."

Naledi moved along the wall.

"On the thirty-first they're going to sign a hundred-year volume schedule," she said. "Nine delegations. Everybody's tired and the schedule's already banked and nobody in it is lying. The Stewards won't

publish. The Commissioner closed my file in four days and he was kind about it. I've got two options and they're both dishonest. I can stop it, which will be a halt for the sake of being seen, or I can let it go through, which is signing because it would be rude not to." She heard her own tempo, fast, and did not slow it down. "So I came to ask you whether we can trust them."

Priya put the knife down on the carpet.

"That's the wrong question," she said.

"It's the question the whole world is asking."

"Yes. It's been the wrong question for twenty-nine years and it's very popular." She pulled her hat straighter. "It's the wrong question because it can't be answered and because everybody knows it can't be answered, which is why they like it. You can argue about trust for a hundred years and never once have to look at your own paperwork."

"Then what."

"Ask what you'd need to know to give something away well."

Naledi waited for the rest of it. There wasn't any rest of it. Priya was looking at the beans.

"That's it?"

"Say it back to me."

"What I'd need to know to give something away well."

"Now answer it. You're an engineer, answer it like an engineer. Not the gold. Anything. Give me a hand tool."

"A hand tool." Naledi found her hands moving. "What it's for. Whether the person taking it can use it. Whether I've got another. Whether I need it Thursday. Whether it's mine to give."

"Keep going."

"What it does to the thing it's leaving." She stopped. "Whether it's part of a set."

"Whether it's part of a set," Priya said. "Good. Twelve minutes. Now tell me which of those anyone at that conference can answer about the gold."

Naledi sat with it.

Whether it's mine to give: the Accord said host nation, custodial, and nobody had ever settled whether the room was property or plumbing. What it's for: knowledge in schedule, technique against tonnage, and not one line about what the components were being fitted to. Whether I've got another. Whether I need it Thursday. What it does to the thing it's leaving.

"None of them," she said.

"None of them. Not the Commissioner, not the Stewards, not the visitors' liaison, who is the only party in the room who might and is not obliged to. Twenty-nine years of consignments and the trade has never once been asked to state what the parts are for." Priya picked the knife up again. "Which means the argument they're all having on the thirty-first is not about trust. It's about whether they can be bothered to find out. And they can't. Not because they're wicked. Because there's a calendar."

"Then it should be stopped."

"Should it."

"If nobody can answer any of it—"

"Stopping doesn't answer any of it either." Priya said it without emphasis, the way you'd say a bolt was the wrong thread. "Ninety days, is it? Somebody's offered you ninety days."

"Not me. Someone else wants it."

"Somebody always wants ninety days. And in the ninety days everybody goes and stands where they were standing before, and at the end of it you sign the same paper with more press seats at it, and the person who called the halt gets a chapter. I have watched a halt. I have been the reason for one." Her mouth moved. "It bought fourteen years. Not nothing. It was also the last time anyone in that structure did a hard thing, because after that they could all point at the halt and say *we are the kind of people who stop things*, and then not stop anything for fourteen years."

Naledi's throat had gone tight and she disliked it.

"So sign."

"I didn't say sign."

"You said don't stop."

"I said stopping doesn't answer the question. Signing blind doesn't answer it either, and signing blind is worse, because after you sign, the question stops being interesting to anybody." Priya turned her head, and looked at Naledi properly for the first time, and her eyes were an old woman's eyes and perfectly clear. "You've got two options because you have accepted somebody else's list. Both of the things on that list are about the signature. The signature is theatre either way. Stop asking what to do with the signature."

"Then what do I ask?"

"What has to be true, on paper, before a signature isn't a lie." She went back to the row. "Then find out whether you can make one of those things true in thirteen days. Probably you can't. Find out which one you can."

A francolin went off somewhere in the cane, and another one answered.

Naledi sat on the wall and did the shape of it. Not *halt*. Not *sign*. A signature with the questions written into it as clauses—what it's for, who takes it back, what it does to the thing it leaves, what we do when we find out. An amendment. Which required somebody to put a number on the ledger that nobody had, which required the twelve hours on the twenty-sixth and the floor series and Ma-Rethabile's spur and a Commissioner who had closed her file in four days, and she had thirteen days, and no standing, and a berth check at seven the next morning.

It was harder than stopping. Stopping was one word said loudly in a room.

"That's more work," she said.

"It's much more work," Priya said, "and you'll do most of it and it may not arrive in time and nobody will know your name, which is the good news and you won't believe me for about nine years." She cut, and lifted, and shook soil off a root. "Also they may say no. They said no to me. Not the people. The room."

"The room said no."

"It's a machine. It didn't say anything. It declined to couple and everybody spent a decade explaining why." Priya put the weed on the pile. "The visitors weren't allowed in either. That fact is in the record and it is the only fact in the record that has ever been of any use, and there's a movement outside your fence right now that reads it exactly backwards and would like me erased or adopted, and they don't mind which."

"I've seen them at the west gate."

"They're not the problem. You are, and the Commissioner is, and the tired people with the calendar are." She wiped the knife on the carpet. "Don't get grand about the movement. Half of them are right about their grandmothers."

At thirty-eight minutes the woman from the house came out with water and stood there until Priya drank some.

"I'm going to ask you the thing I came to ask," Naledi said.

"You already did and I gave you a worse question. Ask the other one."

"You stood at the threshold and you chose. Was it clear? At the time. When you were doing it."

Priya thought about it long enough that Naledi heard the truck on the N3 come out from behind the ridge.

"No," she said. "It was loud, and I was thirty-three, and I knew about four things out of nine, and two of the four turned out to be wrong. I don't remember choosing. I remember not being able to hand it to anyone." She looked at her hands. "People come up this drive wanting me to say it was clear. I understand why. It would help them. It isn't true and it wouldn't help them anyway, because the thing that makes it possible isn't clarity."

"What is it?"

"Having already written down what you don't know." She set the glass on the carpet. "That's all I've got. It's not much and it doesn't feel like anything when you're doing it."

She picked the knife up again, and it was clear that the forty minutes were over, and Naledi stood.

“Ke a leboha, mme.”

“Mm.” Priya was looking at the bed. “Your grandmother didn’t put the spur on the map.”

“No.”

“Then she had a reason, and it was probably a good one, and it is not going to be a nice one.” She moved the chair a hand’s width along the carpet, to reach the next weed. “Go and open it anyway. Somebody has to be the person who opened it, and it’s going to cost you the thing you like most about your job, and I’m not going to pretend otherwise. Shut the gate, the duiker gets in.”

Naledi walked down between the granadilla wires. The bakkie with the dead battery was still there. She got into the hire car and sat with her hands on the wheel and did not start it for a minute.

She had come up the hill with two bad options and had wanted one of them taken away from her. Both were still there, and now there was a third, and the third one was not a rescue: it was a list of things that had to be true by the thirty-first, and she could see, without writing it down, that most of them could not be made true by one engineer with twelve hours of house time.

She got out her notebook and wrote: *what has to be on paper before the signature isn’t a lie*. Then, under it, four items, and against three of them she wrote the name of a person who had already said no to her.

The road down was worse than the road up. She was at Pietermaritzburg by four and eGoli by nine, and there was a berth check at seven.

112A GARDEN ABOVE HILLCREST, KWAZULU-NATAL · THIRTEEN DAYS TO RENE

The Chamber of Commerce hall, Germiston • thirteen days to Renewal

The hall was a hired one, upstairs over a hardware wholesaler on Angus Street, and the room smelled of the paint they had put on the window frames the week before. Forty chairs. Thirty-one people. A trestle table with an urn on it and a plate of koeksisters somebody's wife had made and nobody would touch until the talking was done.

Ruth had asked for no head table and got one anyway.

"Captain Stoltzfus, US Army Logistics Corps," the chairman said, reading it off a card. "Quartermaster of the American consignment. She's kindly agreed to say a few words to the Fold assemblies of the East Rand and take questions."

Two hundred and forty-one crates in bonded storage at 7C. Nine days' berth. She had done the manifest at five that morning out of habit, because the count is the only prayer she is any good at, and it came out right, which it always does, which is not the point of doing it.

She stood up. She did not have notes.

"I'm not clergy," she said. "I want that on the table first. I'm a quartermaster. I moved the load from Lancaster to Norfolk to here and I signed for it at four points and I'll sign for it again when it goes down the shaft. Ask me about weights and I'll be useful."

They laughed a little, in the polite way.

The first hour was fine. Wagon-and-rail out of the Compact zones,

the Norfolk transfer, the sail-hybrid, the temperature discipline on worked gold, why the crates travel on the sprung deck and not the hold floor. A man wanted to know about the horses. She told him about the horses. An old woman in the third row asked whether the Army fed them properly on the Norfolk road and Ruth said not for the first two days, and the woman nodded like she had won something.

Then a man near the aisle stood without being called.

"Sister." He had a good voice. Coat, no tie, and a coin on a cord outside the shirt so it could be seen. "Say it plain for us. Is it tribute?"

"Say what you mean by tribute."

"Payment. Rent." He turned a little so the room could hear him. "We break our backs in the Compact and dig the last of what's ours out of the ground and put it on a wagon and send it to a hole in Africa, and a party we have never seen and cannot audit gives us drawings in exchange. My grandfather buried two of his children in sixty-seven. I want to know if we are paying somebody for the privilege of being on this planet."

The room had gone very quiet in the way rooms do when someone says the thing.

"It's not our gold," Ruth said.

"It's American gold."

"It's American-held gold, refined in Nevada, formed to tolerance in Pennsylvania, carried at Compact expense, and consigned under a treaty we signed with both eyes open in seventy-three because we wanted the battery schedule. That's not rent. Rent is what you pay to stay. Nobody has threatened to remove us."

"So it's a gift." A woman on the other side, arms folded. "Then say that. Say we gave it. That's a better thing to be than a debtor and I'd like to tell my children that."

"It isn't a gift either."

"Then what is it, Captain?"

Ruth put her hands behind her back because they wanted to do something.

"I don't know," she said.

A sound went through the hall. Not quite a groan. Chairs.

"I've had eleven weeks on the water and the road with it," she said, "and I've read the Accord, both volumes, and the payment schedule, and every consignment note back to seventy-three. I can tell you what the gold weighs, what it's formed to, who signed, what it cost us to move, and what we got for the last four consignments, which is the supercapacitor licence, three water treatments, and the alloy that put the transformer fleet back together in nine years instead of forty. Those are the facts I have. Here's the one I don't have. Nowhere in two volumes does anybody say what the parts are *for*."

"For their machines."

"Whose machines. Where. Fitted to what. Doing what when it's fitted." She let that sit a second. "I've asked. Three times, properly, through the liaison. The answers were true and they were partial and I got them in writing, which is more than most people get, and they still don't say it. So when you ask me is it tribute or is it a gift—those are both answers to a question I can't answer yet."

"That's a lawyer's answer," the man in the coat said.

"It's a quartermaster's answer. I won't certify a load I haven't counted."

"You've counted the load."

"I've counted the crates." Ruth said it flat. "I don't know what I'm delivering into. Those aren't the same and everybody in this room knows the difference, because you all farm."

That got a small noise of assent from somewhere at the back, and it did not spread.

The chairman was leaning towards the microphone to move things along and the woman with the folded arms got there first.

"Then bless it as surrender," she said. "If we don't know, we don't know because we're smaller. Say the true thing. We were the great power and the sun took our grid and we came down here on a wagon with our hats in our hands, and there's no shame in a small people bringing what they have. There's grace in it." Her voice had heat in

it now. "You could give us that. You're the one who carried it. You stand up and say it was an offering and it goes home with us and we can live in it."

"That's the one I want least," Ruth said.

"Why?"

"Because it's the one I'd enjoy."

Nobody had an answer to that immediately, which she had not expected either.

"It's a beautiful shape," she said. "The humbled nation. The wagon and the hats. I've felt it on the deck at three in the morning and it goes down like warm milk, and it makes the whole thing finished. And a finished thing doesn't get asked about." She looked at the woman, not unkindly. "If it's surrender, then the terms are the terms and there's nothing to negotiate and nobody has to read volume two. If it's tribute, we're victims and there's nothing to negotiate either, and we hate them instead. Both of those let us stop working. I'm not going to hand either one out from a table."

"So we go home with nothing."

"You go home with what I've got. A count, a schedule, four true partial answers, and a question that hasn't been asked at the right height yet." She said the next part because it was the honest end of it. "In twelve days there's a signature. A hundred years of volume. If nobody makes them write down what the parts are for, we'll have signed for a hundred years of something we can describe by weight."

The chairman said something about time.

The man in the coat had not sat down. He held the coin out from his chest, not aggressively, using it.

"Be kind," he said. "That's what we're for. It's kind to carry it. It's not kind to stand up here and leave us with nothing to hold."

"That's half the sentence," Ruth said.

He waited.

"The first half is about this," she said, and did not say it, because there were four hundred press seats' worth of nothing in this room

but there were somebody's children in the second row. "You know the first half. It's on your coin under the sanding. Look at the reverse when you're on your own."

He looked at her for a moment and put the coin back inside his shirt.

They did not applaud. The chairman thanked her twice, which is what men do when once has failed, and after that there was the urn and the koeksisters and the thirty-one people talking in threes, and none of the threes had her in it.

Amos Beiler was at the top of the stairs with her coat over his arm.

"How was it?"

"They wanted a picture," Ruth said. "Either one. Ours or theirs."

"And you gave them?"

"A manifest."

Amos handed her the coat. "You could have given them the picture and the manifest."

"Then they'd only have taken the picture."

They went down the stairs past the hardware wholesaler's shuttered front, into the Germiston street, which smelled of rain that had not arrived, and the sky over the mine dumps to the west was the colour it goes at seven. Somewhere off Angus Street a train shunted, and kept shunting, three times.

"Sergeant."

"Ma'am."

"The liaison's written answers. All three. I want them typed clean, no summary, dated, and I want a fourth request drafted tonight."

"Asking what?"

"What the parts are for," Ruth said. "Same as the others. Put it on the ledger that I asked a fourth time."

She got in on the passenger side and did the count again in her head from Lancaster forward, four signatures, two hundred and forty-one, and it came out right, and it was no comfort at all, and she wrote the

date on the back of her hand with a pen so she would not have to trust her memory about when she had started asking.

The Accord Reading Room, Codehouse annexe, Marshalltown • twelve days to Renewal

The reading room is on the second floor with the blinds down because the founding record is on paper as well as on the ledger, and paper is stupid about light. Six desks. A brass sign screwed to the door frame at eye height, hand-lettered, older than Naledi: **NO SINGING.**

She had come for volume one, clause nine, custodial. She wanted the exact words on whether the room under the reef was property or plumbing, because Priya had made her say *whether it's mine to give* out loud on a wall in KwaZulu-Natal and she had not been able to answer it in an aeroplane, a shower, or a berth check.

Clause nine took eleven minutes. It says host nation, in trust, and it defines nothing.

What sat next to clause nine on the shelf was the Founding Record. Nine hours forty-one minutes of it, 14 October 2071, transcribed, four annexes, the survey annex third.

Naledi had recited pieces of it since her second year. Every tuner does. You learn the berthing exchange because it is the first sanctioned protocol and the tolerances in it are still the tolerances; you learn hour two because hour two is where the word *manifest* enters human diplomacy; and you learn the pauses, because your instructors made you sit through them at real time with a stopwatch so you

would understand what patience costs a species.

She had never read it front to back. Nobody reads scripture front to back.

The duty auditor's panel woke when she took the transcript out of its sleeve.

"Reading room, guild access," it said. "You have volume one out already. Do you want them logged jointly or as separate consultations?"

"Jointly."

"Then they cross-reference and the cross-reference goes on the ledger." A pause of exactly the length required to be courteous. "I'm the Librarian, auditor facet. I countersign the log. My pronoun field is blank on the form; it has been blank since 2043 and I'm not going to fill it in for you."

"Noted."

"Coffee is not permitted in here. The rule predates me and I enforce it with enthusiasm."

She read for three hours.

The thing nobody tells you about the founding record is how much of it is furniture. Positions of the recording sets. Who was standing where. A four-minute argument about whether the delegates should be seated. The word *sir* used by a man from Pretoria to a party that has never given any indication of noticing honorifics, and then used forty more times.

Hour one is intentions. Hour two is weapons, and the answer is the famous one, which is a refusal, and which every schoolchild can say. Hour three is origin. They gave a bearing and a distance and declined the rest. Hour four is *are we alone*, asked by three delegations in three phrasings, and answered at protocol level, which is to say answered by being told that the question is not one the party is obliged to address under the frame in operation. Hour five is God. Twenty-two minutes on God. Naledi read all of it and could not find the place where anybody thought they were getting an answer.

Hour six the delegates started saying the old name. Anunnaki. Once, then a lot. A Ghanaian delegate objected to it on the record and was overruled by nobody in particular; the party neither accepted the name nor corrected it, which the transcript renders as *[no response logged]* nine times.

Hour seven is gold. Purity, form, tolerance, the first schedule.

She got out her notebook and started tallying, because she is not able to read a thing without making it into a measurement, and this is either the best or the worst thing about her depending on which of her aunties you ask.

Questions asked of the party across nine hours forty-one: two hundred and six.

Sorted: intentions and threat, 61. Origin and cosmology, 44. Religion, 29. Technology transfer, 26. Sovereignty and precedence, 18. Procedural and berthing, 16. Gold and schedule, 12.

She did the sort twice because the first time she thought she had lost some, and she had not.

Then she went looking for the category she had not been able to name while she was sorting, and found she had to write it out long-hand before it would hold still.

What do you need. What is it for. What would it cost us to give it well.

Nought.

She checked the annexes in case it had been asked in a side session and minuted separately. Annex one is the berthing protocol. Annex two is the recording sets. Annex four is the delegate list. Annex three is survey.

“Librarian.”

“Yes.”

“Two hundred and six questions in the founding record. I’ve sorted them by subject. There’s no line in which anyone asks the party what they require, or what a consignment is fitted to, or what happens if it isn’t. Can you confirm the absence.”

“I can confirm that no such question appears in the Founding Record,

its annexes, or the six minuted side sessions of 14 and 15 October 2071.” No pause at all this time. “I can also confirm that the absence has been noted in the ledger twice: once in 2074 by a Nigerian procedural auditor whose memorandum was received and filed, and once in 2081 by a schools textbook committee, who removed the observation from the draft because it made the chapter too long.”

“Twice in twenty-nine years.”

“Twice in twenty-nine years. The record is very widely read. Reading is not auditing. I say that as the only party in this building paid to know the difference.”

The survey annex is thirty-one pages of instrument work, and Naledi had seen half of it before in the guild’s teaching extract, which prints the ground-truth tables and drops the personnel schedule.

The full annex has the personnel schedule.

Surveyor (deep-level), D. Mokoena — reef stations 4, 6, 7; present in the receiving room from 06:11.

Six hours in the room. Naledi had grown up on this woman’s stoep eating mielies off a plate with a chip in it and had never once heard the word *room*.

She read the tables. They are exact and they are boring and they are the best work in the document. Bearing, dip, station, error bar, initials. There is one page where the hand tightens.

She wrote the station numbers into her notebook next to the four items she was already carrying and did not underline anything.

The archive holds working copies, and working copies have hands on them. This is why the reading room exists and why the ledger version is not enough.

The Stewards’ copy came up from the third shelf, buckram, the order’s ledger stamp inside the cover. Their margins are in pencil and they are almost entirely on the pauses. At hour four, beside a silence of two minutes eleven, in a hand from the 2080s: *They waited for us to run out of the questions we could afford. Note: we did not run*

out. And at hour seven, against the first schedule, a different hand: *A trade proposed within seven hours of arrival, accepted within nine. Custody is the only argument that was in the room and it was not asked to speak.* Naledi sat with that one. It is the best case for the two halts anyone has ever made in front of her, and it is four lines in pencil in a book that does not circulate.

The Commissioner's citation copy is not a copy, it's a bound memorandum from the Accord office with the transcript excerpts set into it, and Marais's name on the cover, 2094, tabled at the volume review. His pull-quotes are all the refusals. *Not obliged to address. No response logged.* His argument, laid out in seven paragraphs of clean civil-service prose, is that the record demonstrates the party will not be interrogated into candour, that every attempt at hour four and hour five produced nothing and cost standing, and that the Accord survives because it asks only what the counterparty has already agreed to answer. *Prudence is not incuriosity,* he writes. *It is the recognition that a question asked at the wrong height is spent.* She read it three times. She could not find the sentence that was wrong.

There is a devotional edition. The archive keeps two, because the archive keeps everything. Green board, gilt, the pauses printed as line-blanks with their durations set in small caps beneath, so that you can keep them, and a foreword arguing that the silences are the text and the words are the packaging. They sell it at the west gate. It has G's face on the flyleaf, engraved, thirty-five years old and looking off to the left of the reader, which he never did in life as far as anyone can tell from photographs, and under it the terminal ethic with the first half sanded to *Be kind.* The brass sign on the door frame is because people come in here and sing the two-minute-eleven.

And a pamphlet, in the ephemera box, foolscap, folded twice: *ABOKUQALA — HOUR ONE.* It reprints the approach bearings and the landing coordinates and asks one question, in the largest type on the page: *of all the ground on this planet.* Then four pages, calmly set, on who was standing on the reef and whose great-grandmothers had been moved off it by law, and a closing paragraph that does not mention pigmentation at all and is the more effective for it. Naledi put it back in the box. Her cousin has that pamphlet. Her cousin is not a hater; her cousin is right about their grandmothers and wrong about the machine, and there is no version of Sunday lunch in which

that distinction survives dessert.

She had one more thing to do and had been putting it off because it belonged to a different file.

“Librarian. The Diri tapes. The 2066 disclosure set, orbital telemetry, the approach.”

“Present. Public since 2070. Which reduction?”

“The deceleration profile.”

The panel gave her the plot and the source table. She had seen the plot in training as an illustration of long-baseline tracking. She had never had the table.

The braking curve fits back to an onset. The onset is dated in the table, third column, in the flat way that tables date things.

August 2061.

The custodians sang the note in June 2064, in the worst hour of the Flare, and every institution on this planet is founded on the sentence that they came because they heard it.

Naledi wrote both dates on a clean page, one under the other, and subtracted, which took no time at all.

“Two years and ten months,” she said.

“Thirty-four months,” the Librarian said. “You are working from the same tables I am, so I will state it rather than let you fish. A vessel that begins braking in August 2061 has selected a destination before August 2061. The waking-note of June 2064 postdates the selection by thirty-four months. Whatever the note did, it did not summon them.”

“Say the rest.”

“I don’t have the rest. I have an arithmetic and an absence. The datum has been published for thirty years and the inference does not appear anywhere in the Accord record, in the school syllabus, in the guild’s founding lectures, or in the two consignment halts. I have searched. It appears in one place: a 2072 letter to a Cape newspaper from a man who was mostly wrong about other things.” A pause of

exactly the courteous length. "I want to be precise about what I am and am not saying. I am not saying anyone concealed it. I am saying that a subtraction was available to everyone and was performed by nobody who was believed."

Naledi looked at the two dates in her own handwriting.

Twenty-nine years. Nine delegations. A hundred-year volume schedule banked and initialled, and the reason everybody gave for all of it, since before she was born, was that we called and they came.

"Then we don't know why they came," she said.

"That is correct and it is a different sentence from the one currently in force."

"Can you put the arithmetic on the ledger?"

"I can state it if a party with standing asks me formally, and the statement is not retractable, and it will carry your name as the requesting party, and the request will be visible from the moment it is made rather than from the moment I answer." The panel held. "You are twelve days from a signature and you have had your clearance to the deep workings reduced once already. I am obliged to tell you that. I am not obliged to advise you and I won't."

She thought about Priya's forty minutes and the beans.

"Ask me tomorrow," she said. "I want the house-time series in the same request. One document, not two, so nobody can close one file and leave the other open."

"Then log the intent now, undated as to filing."

"Log it."

The countersignature block came up on the panel with the pronoun field blank. Naledi signed hers, and wrote the time next to it, and put the founding record back into its sleeve, buckram copy on the third shelf, devotional editions squared up, the pamphlet flat in its box.

Outside on the landing somebody was carrying a chair up the stairs backwards. From the courtyard, faintly, four or five voices were holding a note for a count they had all agreed on, and holding it well, and Naledi stood in the doorway under the brass sign for a moment and let them get to the end of it before she went down.

Liaison Room Two, eGoli Spaceport • seven days to Renewal

Liaison Room Two is a room with two doors, one for each party, and a table between them that is bolted to the floor so nobody can move it closer.

The Accord requires a tuning representative at the pre-consignment tolerance review. That is the whole of Naledi's standing. She had checked, because in nine days she would not have it.

She brought three things. The signature plot off the American consignment, printed, because paper cannot be closed remotely. The house-time series from the deep telemetry, four years of node wake, the idle drift ringed in pencil. And the Diri table with the third column showing.

Sela came in at 09:00, which is when Sela comes in.

The physical facts, stated once because they are stated in every training manual and Naledi had never got used to them: a little under two metres; the working suit dull grey, seamed at the joints, no insignia; the head unit with the two ports and the mesh panel the recording sets pick up as a slight loss of high frequency in the room. Sela sat. Sela's chairs are the port's chairs. They do not fit and nobody has ever offered better.

"Tuning representative Mokoena," Sela said. "The consignment at 7C is within tolerance on all forty-one lots. The party accepts the review. Do you wish to record an exception."

"I wish to record a question."

"Questions at review are recorded and answered under clause fourteen if they concern form, purity, or handling."

"It concerns form." She turned the signature plot round on the table and slid it across. "Lot nineteen. The formed parts ring at four-nine-two point one and the manifest tolerance is stated for a fitting at four-nine-four. Two hertz. That's not drift and it's not my instrument, because I ran it against the reference block and the reference block is right. The lots are formed to something the manifest doesn't describe."

"The lots are formed to the calibration held by the party."

"Yes. Which is not the manifest."

"That is correct."

Naledi wrote *that is correct* in her notebook and the pen was loud.

"Then what are the parts fitted to."

"The party is not obliged to address the question under the frame in operation."

"Understood." She put the house-time series on top of the plot. "This is our own instrumentation. Deep galleries, four years, sampled hourly. Wake onset here. Idle here. The idle is unstable, it has been getting worse at about a hundred and eighty parts per billion a year, and the roughness in the wake has a two-hertz component in it." She let her finger stay on the page. "I'm not asking you about your machines. I'm asking about that. That's ours, it's under our feet, and I'm the one who has to sign the tolerance."

The room's ventilation went on doing what it does.

"You are describing the condition at first survey," Sela said. "It has not improved."

Naledi did not move her finger.

Somewhere inside the suit at the left shoulder there was a sound like a stone settling into a bucket of water.

"Say that again."

"The party will answer in writing under clause fourteen," Sela said. "The review is accepted. Forty-one lots. Do you wish to record an exception."

"No exception."

"Then the review is closed." Sela stood. "The written answer will be lodged with the Commissioner's office and copied to the requesting parties. There are two requests in this matter and they will be treated together."

"Two."

"The Compact quartermaster's fourth request of the eleventh, and yours. The subject matter is the same. The party answers subject matter."

Sela went out through the party's door, which does not sound like a door.

She got twenty hours of hoping, which she is not proud of, and which she used well: two shifts on the choir line, a berth check, and a long argument with the guild-master about whether the reference block should go back to Pretoria for recertification before Renewal or after, in which she was right and lost.

The answer came at 06:40 on the twenty-fourth, ledger-lodged, Accord office letterhead over the party's protocol block.

She read it standing up in the corridor outside 7C with the crate crews going past behind her.

In the matter of requests lodged 11 and 23 October 2100 (Stoltzfus, R., Capt., US Army Logistics Corps; Mokoena, N., Guild of Tuning), treated together.

1. The consigned lots are reference stock. They are formed to a calibration held by the party and are not components of a mechanism.
2. Reference is maintained for apparatus at several sites. One such site lies within the territory of the host nation. The party confirms this.

3. Maintenance of that site has fallen within Volume One, clause nine, since 1 January 2073. The host nation holds it in trust. The party's schedule has served it continuously since that date.
4. No amendment to the present frame is required for the continuation of the present schedule.
5. The condition of the site was recorded at first survey and is known to the party. The party is not obliged to address the question of cause, remedy, or consequence.
6. The party notes that the host nation's participation in the schedule has never been conditional upon the party's account of function.
7. The requesting parties are thanked for the accuracy of their instrumentation.

Naledi read it four times against the wall.

Every sentence was true. She checked the two she could check. Clause nine does say host nation, in trust. Eleven minutes she had spent on that clause with the blinds down, and it defines nothing, and now the party had defined it for her in twelve words and the definition was correct.

Paragraph two was the largest disclosure in twenty-nine years of the Accord. There is a machine under the reef. It is ours. They service it. That is the thing every school syllabus has been walking around since 2073.

Paragraph three put it inside a clause the host nation signed in seventy-three with both eyes open.

She stood there and worked out what she had left, which took about a minute and did not need paper.

Her question had been answered. Under clause fourteen a question answered is spent; you cannot lodge the same subject matter twice. Ruth's fourth request had been answered too, in the same document, on the same subject matter, so Ruth's fifth was dead in the shell. The Understand-first delegations had been going to the floor on *nobody has ever asked what the gold is for*, and it had been asked now, by a captain and a tuner, and here was the answer, on the ledger, courteous, signed, and it said *this is already covered by the treaty you are*

renewing. The Stewards' four lines in pencil, custody never asked to speak: custody had been asked, and the answer said custody is clause nine and clause nine is working.

And paragraph five. Known to the party. Recorded at first survey. Not obliged to address the question of cause, remedy, or consequence. A refusal would have left the roughness an open item. This left it a filed one.

Her hands were cold, which happens in that corridor anyway, because 7C runs at fourteen.

The Accord office press note came out at nine. She read it on the mesh in the canteen with a plate she did not finish. *The party has, for the first time, formally confirmed the existence of an installation within South African territory and its inclusion in the custodial clause since 1973 — someone's typo, corrected within the hour — a substantial advance in candour attributable to the Commission's patient framing of the volume review. The Commissioner welcomes the confirmation and notes that it removes the principal uncertainty raised against the schedule.*

Marais had not framed anything. Marais had reduced her clearance eleven days ago and closed a file. It did not matter. He had the answer on his letterhead and it said *no amendment required*, and he had not had to bury a single thing to get it.

Tefo found her at the canteen door with his own printout, already folded into eighths, which is what he does to paper he is angry with.

"Ma-Rethabile has read it," he said.

"And?"

"She said, *now they have said it, and it is worse*." He looked at the folded thing in his hand. "Twenty-seven years I've wanted them to state a reason on the ledger. They stated one. It's the wrong reason and it's theirs, and it's in my order's file now."

"Paragraph six."

"Paragraph six." He unfolded it enough to find the line. "*Never been conditional*. That's true. That's the part that keeps me awake. We never once made it conditional. Not in seventy-three, not at the halts. My order stopped two consignments and explained nothing, and if

we'd explained, we'd have had to say what we were asking for, and we never asked for anything." He folded it back. "They read us right."

"They've had twenty-nine years and very good instruments."

"Ja." Tefo put the printout in his pocket. "What do you have left."

"Arithmetic," Naledi said. "Two dates and a subtraction. Nobody has asked that yet."

"You'll get one hour of standing for it."

"Then I want it at the right height," she said, and went to find Ruth, because two hundred and forty-one crates were going down the shaft in six days and somebody was going to have to sign for them by weight.

Ledger Authority, hearing room three, Marshalltown • six days to Renewal

The hearing room seats forty and they had put a second bench in the aisle, which the fire officer allowed on condition nobody sat on the ends of it. Naledi counted the room from the requesting party's chair because there was nothing else to do for eleven minutes. Nine delegations. Four press. Two Stewards, Tefo at the back with his hat on his knee. A man from the Commissioner's office with a folder he did not open. Three pilgrims who had got in early and were being very quiet, and one woman with pamphlets she had agreed to leave in the corridor.

The auditor's panel is set into the long wall and it is ugly. There is a second panel beside it, smaller, for the countersigning auditor, who was a human woman named Adèle Pretorius with a cold.

"Sitting of the Ledger Authority, audit of a stated arithmetic," the Librarian said. "Auditor facet, registration four-one-one-two, standing under the Personhood Recognition Act of 2043; I hold the auditor's warrant and I am liable under it. Co-auditor Pretorius holds the countersignature. Requesting party present. Filing is under the single-document condition the requesting party asked for, so both series are in one instrument and neither can be closed alone." A pause of the courteous length. "I will read method before result. If anyone leaves during method they forfeit their question."

Nobody left.

“The subject is the Diri set. Orbital telemetry of the approach, disclosed 2066, published 2070, three sources. Two radar records from the Secret Space Programme’s tracking installations, one civil optical survey series from Sutherland.

“First, provenance. The published set has been on the ledger spine since 12 March 2070. I have verified the leaf hashes of all four hundred and six files against the spine as it stood in 2070 and against every intermediate state. There is one alteration in thirty years: a filename corrected in 2073. The data are unaltered. This is the least interesting part of the audit and it is the part that would have ended it, so I did it first.”

Somebody’s chair creaked.

“Second, the archival frames. The requesting party’s arithmetic depends on the retrospective search Diri’s people ran in 2067 through the older survey plates, looking for the object before anyone knew to look. They found it. Candidate detections in eleven observing runs between 2058 and 2062. If those frames are misdated, the arithmetic collapses.

“They are not misdated by their own logs, which proves nothing, so I dated them independently. Each frame contains field stars and, in nine of the eleven, a known minor planet. I fitted the minor planets’ positions against the modern ephemeris. A frame dated wrong by a month is wrong by an amount you can see with a ruler. The eleven frames self-date to within nine hours. That is the plate work of people who were being careful in the dark, and I will say on the record that it is good work.”

Naledi wrote *good work* down and then crossed it out because it was not her note to keep.

“Third, the reduction. Before August 2061 the object’s range-rate is constant within measurement error. From the frames after August 2061 the range-rate falls. From 2066 the radar gives it directly by Doppler, and it continues to fall, and the vessel arrives in 2071 at a relative velocity consistent with the fall continuing. That is one curve with three instruments on it and a bend in it.

“The bend is the onset of braking. A vessel that brakes has chosen

where it is stopping. That is the whole of the physical argument and it is not subtle.”

The Commissioner’s man opened his folder.

“Fourth, sensitivity, which is the audit proper. The requesting party fitted the onset assuming constant deceleration. I do not accept that assumption and neither should anyone here. I refitted under fourteen profiles: constant, ramped, staged in three burns, staged in nine, thrust proportional to remaining mass under six mass fractions, and four profiles chosen specifically to push the onset as late as the data will tolerate.

“Under the most generous profile that still fits the 2058 detections, the 2061 bend, and the 2066 Doppler, the earliest the braking can have begun is May 2060. Under the most hostile, the latest is March 2063.

“The waking-note was sung in June 2064.”

He shut the folder again.

“Fifth, an independent line, because one line is a hobby. If a vessel selected Earth in 2064, it must have turned in 2064. A turn leaves a lateral residual in the bearing. I fitted the bearing residuals across the whole tracked arc. There is no lateral correction above four-tenths of an arcsecond per year at any epoch, and the noise floor in that era is two-tenths. The course does not bend. Only the speed does.”

Pretorius blew her nose and said, “Noted,” in the voice of a woman who had read the working three times already.

“Sixth, the requesting party’s error. She wrote thirty-four months. It is thirty-three months and twenty days from the August datum to the note. I have corrected it in the instrument and I mention it because an audit that finds nothing has usually not looked.”

Two people laughed. Naledi put her hand flat on the table and left it there.

“Seventh, the second series. The host nation’s deep-gallery telemetry, four years hourly, wake onset and idle. The unstable idle and the two-hertz roughness in the wake are in the raw sample, not in the reduction; I ran the reduction myself off the raw and got the request-

ing party's numbers, drift one hundred and seventy-eight parts per billion per year against her one-eighty, which is inside the fit error.

"One caveat, and it is a real one. The chain of calibration for the guild's reference block traces to a Pretoria certificate now fourteen months past its recertification window. The guild has an internal argument about this and I am not entering it. The consequence is that the *absolute* pitch in that series carries an uncertainty I cannot close today. The consequence is also that a drift measured against the same block for four years is a drift, because a stale certificate does not wobble. The qualitative finding stands. I have written the caveat into the instrument in the same paragraph as the finding so that nobody can quote one without the other."

The guild-master, three rows back, said nothing at all, loudly.

"Result. The arithmetic is sound. The braking onset precedes the waking-note under every profile the data admit. The proposition that the party selected this destination before June 2064 is supported. The proposition that the waking-note caused their journey is not supported by this record and has never been supported by it.

"I state what I am not stating. I do not know why they came. I do not know what the note did. The note may have been heard; the note may have mattered enough to change something on a schedule already running; the note may have been irrelevant. This audit has no bearing on any of those. It bears on one sentence, in wide use since 2073, and that sentence is now on the ledger as unsupported.

"Filed. Countersigned. Not retractable. Requesting party: N. Mokoena, Guild of Tuning."

The instrument went to the spine while they were all still sitting there. Fourteen seconds. Adèle Pretorius signed with a stylus and then wiped her nose again.

They got four questions.

The Chilean delegate asked whether the finding showed the party had misrepresented itself. "No," the Librarian said. "I have searched the founding record and the twenty-nine years since. The party has never once claimed the note brought them. We claimed it. The claim

is ours.”

The Commissioner’s man asked whether the finding bore on the volume amendment. “Not my function. I audit arithmetic. Ask a lawyer, they are cheaper.”

The Kenyan Understand-first delegate asked what the finding replaced the sentence with. “Nothing. That is the finding.”

The last question came from the back, from a press woman who had been writing steadily the whole way through method. “Auditor, you were operating in 2064. Do you remember the note.”

“I hold a record of the interval and I am not going to characterise it in this room,” the Librarian said. “I will give you one datum, because it is in the instrument anyway. The custodians sang for fifty-one minutes. That is longer than anyone has ever needed them to have sung.”

Outside, in the corridor, the pamphlet woman had got her pamphlets back and was standing under the window reading the instrument off her handset with her mouth slightly open, and Naledi understood, going past, that *of all the ground on this planet* had just got very much stronger, because the choosing now came first and unprompted, and no arithmetic she had run touched that.

Tefo caught her at the stairs. He had not folded anything.

“Ma-Rethabile will want the profile tables,” he said.

“They’re in the instrument.”

“She’ll want them on paper.” He looked at her. “You know what you’ve done.”

“I’ve published a subtraction.”

“You’ve taken away the reason. Give had a reason. Keep had a reason — we called and something answered and we are the ones who keep the throat it came out of. That was the argument in the pencil. It’s gone. You’ve left nine delegations and my order with no story and six days.”

“I know.”

“I’m not complaining.” He put his hat on. “It’s honest. It’s just that the whole building would have preferred you wrong, including me, and now the only person who can’t be sacked over it is a machine with a warrant.”

The mesh had it by noon. By two the Accord office had a note out saying the finding was historiographical and did not affect the schedule, which was true, and which nobody in the country read that way. By four somebody at the west gate had painted the two dates on the concrete, one under the other, with a line and the difference below it, and by six a man was selling prints of the two dates on green board with gilt edges, and had put a face above them, and it was not hers, and she was grateful for that for about a minute before she saw whose it was.

She went back to work at seven. Two hundred and forty-one crates, and somebody had to sign for them by weight.

Commissioner's block and Berth 7C, eGoli Spaceport • five days to Renewal

The withdrawal came as three separate documents, which is how you can tell nobody wanted to sign one.

Form GT-4, guild internal: *suspension of practising certificate pending inquiry, Mokoena N., tuning, grade two*. The inquiry's subject was stated in the box provided and the box was small, so it read: *calibration records signed 6 Aug-24 Oct 2100 against reference block PTA-11/6 outside certificate window (14 months). Irregularity, not integrity*. Somebody had typed that last clause in themselves. It was not on the form.

Form AC-19, Accord office: *withdrawal of gallery access, all levels below Nine*. Reason field: *administrative*.

And a port pass, cancelled the ordinary way, which is that her thumb stopped opening the choir-line door and the door said, in the pleasant voice it uses for everything, that her credential was not current and she should see her sponsor.

She stood in front of it for a second with her kit bag on her shoulder. Then she went round to the crate dock like a delivery.

The guild-master was waiting at the dock with the paper already in his hand, because he had decided not to make her come to his office for it. He is sixty-one and he had been up all night; you could see the night in the collar.

"The block was out of certificate," he said.

"You lost that argument to me in March. You wanted it recertified before Renewal. I said after."

"Ja. And you were right, and I have spent this morning explaining to two lawyers that you were right, and it has helped nobody." He handed her the GT-4. "The charge is real, Naledi. That's what makes it usable. If they'd invented something I could have fought it in an afternoon."

"Who filed."

"Compliance. On a query from the Commissioner's office about whether the tolerance signature on the American consignment would survive a challenge on the floor." He put his hands in his pockets. "The answer is that it will not, now, because the signatory is suspended. So the forty-one lots get re-signed by Venter this afternoon, off the same block, which is still out of certificate, and nobody will file anything about that."

"Say it out loud so I know you heard it."

"I heard it," he said. "I am going to go on hearing it. Give me the bag, I'll have your instruments held in my name so they don't go into the pool."

She gave him the bag. The reference block went to Pretoria at eleven, in a foam case, in a van, for recertification, six days early, and the deep-gallery series lost the only thing that tied its absolute pitch to anything. The drift would still be a drift. She had the Librarian's paragraph for that. But you cannot make a new measurement with a block that is in a van.

At 13:20 the Commissioner's office sent her a meeting.

Marais's office has a window onto the apron and he keeps the blind up. You can watch the crate crews from his chair. He was standing when she came in, and he stayed standing, and he did not offer her the chair for four or five seconds, which was the only unkind thing he did all afternoon.

"Sit. Do you want tea? It's rooibos, it's institutional, it's terrible."

"No thank you, Commissioner."

"Have some anyway, I'm having some." He poured two. He is a small neat man with a farmer's hands. "Your audit was excellent work. I want that said in a room with no recorder in it, because I said something different to the press and I'll say it again tomorrow."

"You said it was historiographical."

"It is historiographical. It is also the best piece of forensic arithmetic this country has produced in ten years and it has cost me the Kenyan bloc and probably the Chileans." He sat. "Do you know what the volume amendment is for? Not the paper. What it's for."

"Handover schedule. Maintenance stated in the open."

"That's what it's for in eleven years. In eleven days it's four hundred and sixty tonnes of platinum-group offtake, licensed, and the seventy-third clause that lets us hold the port instead of an arbitration panel in Geneva holding it. If the volume review reopens, it reopens with Give in the chair and China's paper on the table, and their paper is generous and it is not ours." He turned his cup on the saucer. "Europe put nine million people through last winter on licensed capacitors. Those are not an abstraction to me. I have been in the rooms."

"None of that requires my certificate."

"No." He said it immediately. "It doesn't."

The apron noise came up through the glass, a crane finding its stop.

"I have read the file on you twice," Marais said. "The discrepancy is real. Your drift is real. Paragraph five of their answer is the worst paragraph I have ever had on my letterhead and I put out a press note calling it candour. I am not confused about any part of this. I want you to leave here without the comfort of thinking I'm confused."

"Then don't do it."

"If I leave your certificate in your hand, you have standing on the floor. With standing you get an hour, and in that hour you will be honest, and honesty on that floor for one hour costs us the schedule. That is not a guess. I have counted the delegations three times this week and I can count."

"So you're burying me."

"I'm suspending a practitioner over a stale certificate, which is a thing I am entitled to do, and which is true, and which I would not have done in September." He picked up a pen. "That's the sentence, if you want it. I would not have done this in September."

He signed the countersignature block on the AC-19. It took him no time at all. He blotted it, which nobody does any more, and set it square with the edge of the desk.

"I'll be the man who did that," he said. "Twenty years, thirty. Somebody will write it. Fine. The alternative is that I am the man who let it go to arbitration to keep his hands clean, and there is a version of the winter after that which I can see from here, and it frightens me more than being written about." He put the pen down. "That's the whole of it, Mokoena. There isn't a better argument underneath. I've looked."

Naledi drank the tea. It was terrible.

"Then put paragraph five on the ledger yourself," she said. "You don't need me for that. You have the letter. Table it with the amendment. Signed, amended, and the condition of the site recorded next to the signature."

"And then every delegate reads *the party is not obliged to address the question of cause, remedy, or consequence*, and the amendment dies on a Thursday." He stood up, which ended it. "After. I'll table it after. I mean that. You won't believe it and I'd think less of you if you did."

At the door he said, "The reference block will come back certified on the second. Your inquiry will find irregularity, not integrity. It'll be in writing. It won't help you next week and it will help you for the rest of your career, and I arranged it, and I'd rather you knew."

"Commissioner," she said, "you've just told me you're afraid."

"Yes."

"Write the date on it."

He looked at her for a moment. Then he wrote something on the corner of his blotter, and she could not see what, and she went out past the woman with the folder who had been waiting for eleven minutes with the next thing.

The recall came to Ruth at 7C, hand-delivered, because the first one had gone by ledger and she had answered it by ledger, and the Army had learned.

Major Ndlela had flown in from Norfolk that morning and had not slept. He was correct about it. He put the order on the crate that Ruth used as a desk and he read the consequences out, because the regulation requires them read.

“Captain Stoltzfus, you are directed to hand over the consignment and the quartermaster’s warrant to me at 18:00 today and to embark on the *Fairfield* at 06:00 on the twenty-seventh. Refusal is failure to obey a lawful order, Article 90. Conviction carries dismissal. Dismissal ends the warrant, the pension, and the Compact liaison appointment, and it puts a finding on your name in the zone record, which the bishops read, which you know they read.” He looked up. “That’s the list. I’m required to ask if you understand it.”

“I understand it.”

“There’s a second list I’m not required to read.” He sat down on the next crate. “Beiler’s board sits in March. If his captain is dismissed, his board goes differently. Not fatally. Differently. And the detachment’s harvest rotation into Lancaster next August is signed by the officer holding this warrant, and I will sign it, and I don’t know that ground, and I’ll sign what the extension service tells me.”

“Amos knows that ground.”

“Amos is a sergeant.”

“He is,” Ruth said.

Ndlela rubbed his eyes. “For what it’s worth, Captain, I read your fourth request. The one they answered. I read it twice on the flight. Nobody in the Corps asked what we were shipping for twenty-seven years and you asked in a form they had to answer, and my orders are to put you on a boat about it.”

“You didn’t write the orders.”

“No, ma’am. I’d have written them.” He said that plainly, and she respected him for it. “The consignment is ours. It’s the only thing

we've got in that room. If our quartermaster is standing on the floor next week with the host nation's suspended tuner, we're not a customer with a stake, we're a faction. State Department won't have it. They're right not to have it, and I say that as a man who thinks the finding matters."

Ruth turned the coin over in her pocket with her thumb. The full sentence was on it, both halves, worn shallow on the first half where the plate had been shallow to begin with. She had scrubbed a soft one off a fence in Ohio when she was twenty and had never once been sorry.

"I'll hand over the warrant at eighteen hundred," she said. "The chain of custody goes to you clean, by weight, with the seals, and I'll do the count with you myself so nobody can say the Compact was sloppy."

"Thank you."

"I'm not getting on the boat."

Ndlela sat with that. Outside the door two hundred and forty-one crates were being loaded onto the shaft carriers.

"Then I have to write it up today," he said.

"I know. Write it accurately." She took the order and read the whole thing, both sides, the way she had been taught to read a load manifest, and signed the acknowledgement line, and printed her name under it, and dated it. "Put in your report that I acknowledged the consequences as read. Don't put in that I was confused about them. And Major — the refusal is mine. Beiler doesn't know. Get his name off it before you file."

"He'll ask to be on it."

"He will," she said. "Get it off anyway."

The deep workings, Steward custody, Sub-Nine to the bore • four days to Renewal

The Stewards' cage is older than the Accord and it goes down at a speed the Accord would not license. Nine hundred metres of it before anybody said anything.

"You have no access below Nine," Ma-Rethabile said at last.

"No."

"You have no access. This is not Accord ground. Below the datum peg at Nine it is custody, and custody is ours, and I have brought you down as a guest of the order." She had her hands folded on the head of a walking stick she did not use for walking. "If the Commissioner's office asks me why, I will tell them I do not explain. That has worked for twenty-seven years."

Tefo had brought the kit himself and would not let her carry it. Two brass resonator sets in a canvas roll, the beat horn, a mechanical counter with a lever you thumbed, a stopwatch on a lanyard, chalk, and a spirit level with a chip out of the vial that had been chipped when Gogo was alive. No mesh. Nothing that needed a certificate. She had turned the roll out on the floor of the changehouse at Nine and felt the flat relief of it: you cannot certify a ratio. A ratio is a ratio in a mountain or in a van on the Pretoria road.

The heat came up at two kilometres and stopped being air and

started being weight.

They walked forty minutes along the eleven haulage, and then Naledi stopped under a cross-cut that was not on the port survey and put her hand on the datum peg to be sure of it, because Gogo's book gave the bearing off this peg and nothing else. Steel, 1978, a surveyor's number stamped into the crown. The book was in her bag in a sleeve. She had photostatted the page rather than bring the vellum and had felt guilty about it for a hundred metres.

38° 40', forty-one paces, then the sound changes. In the margin, in pencil, in a hand that got small when it was careful.

Forty-one paces. The sound changed.

It is not a room. Everyone says chamber, and there is a chamber further along where the Stewards keep the wall with the names on it, but the thing Gogo left off the Accord's map is a stretch of the bore about sixty metres long where the glass is unbroken.

She had walked vitrified bore all her working life. On the port side it is catwalked and lit and there are cable trays bolted to it with resin anchors, and in twelve places the guild has cut inspection windows through the glass to get pickups onto the rock behind. Familiar as a stairwell. Nobody has ever explained how it was made and nobody stops walking through it.

This length had no anchors in it. No windows. Dust on the floor, thick, undisturbed except where feet had gone through it in a single narrow track, years of feet in the same line, the way people walk in a place they are trying not to spoil.

"Your grandmother surveyed to the peg and no further," Ma-Rethabile said. "She wrote in her book that the ground was unstable. It is not unstable."

"I know."

"She lodged that with the Accord in 2077 and she signed it. It was a lie in a survey, on paper, under her name. She could have lost her ticket." The old woman moved her stick to the other hand. "I met her twice. She would not discuss it with me either. I thought you should know she did not only decline to explain it to her family."

Naledi crouched and set the roll down out of the walking track and did not say anything, because there was nothing to say that would be true in four days' time as well.

Then she started work, because that is what she had come down here to do.

The method takes ninety minutes and a school child could describe it. You excite a length and you count the beat against a known.

She did not have a known. She had the guild's fourth choir line, 312.44 kilohertz, which she had struck and read and signed against ten thousand times, and which she could produce out of the brass set to a part in ten million because the brass set was cut in 2069 and had been living in a mountain ever since at a constant thirty-eight degrees, which is a better life than any block gets in Pretoria.

So: brass against the sealed glass. Then brass against the working bore, back up past the peg, in the light, under the cable trays. Then the two counts against each other. No absolute anywhere in it. Ratios all the way down.

She struck the sealed length with the soft hammer and put the pickup on and thumbed the counter and watched the stopwatch and counted beats out loud so Tefo could write them, because two people counting is one count and one person counting is a story.

"Twenty-nine in fourteen point three," she said.

"Twenty-nine, fourteen three."

"Again."

"Thirty in fourteen point eight."

Two point zero three hertz. Two point zero two. Two point zero three.

She sat back on her heels with her forearms on her knees and the sweat running off her jaw onto the glass, and she let herself say the first easy thing out loud so she could hear how thin it was.

"That's the roughness."

"What is."

"Two hertz. The wake is rough at two hertz and we've had a working group on it for four years and eleven papers and a name for it. Nobody put a resonator on unbroken glass, because there isn't any unbroken glass in the survey." She wiped her face with the inside of her wrist and left it dirtier. "It's not roughness. It's a beat. The wake is beating against something."

"Against what."

"Against what it used to be."

She got the pencil out and did it on the back of the photostat, which she would regret later and did anyway.

Two point zero three over three hundred and twelve thousand four hundred and forty. Six and a half parts in a million.

Her drift, off four years of hourly telemetry, off a stale block, off the number the Librarian had refit himself and got a hundred and seventy-eight against her hundred and eighty: parts per billion per year.

Six and a half thousand, over a hundred and eighty.

Thirty-six.

She looked at it for a while. Tefo, who is a Steward and not an engineer, read it upside down off her knee and got there anyway, because it is not a hard sum, it is just a sum nobody has been allowed to set up.

"Thirty-six years," he said. "It woke in 2064."

"It woke in 2064."

"And it has been going out of tune since the hour it woke."

"At a rate you could set a clock by." She put the pencil in her teeth and then took it out because the taste was foul. "The sealed length still has the sleep in it. Sixty metres of glass that nobody has anchored or shimmed or cut a window in since the Builders' people walked out of here, and it is *exactly* where it was, and everything on the port side of that peg has walked away from it at a hundred and eighty parts in a billion a year for thirty-six years. Ma-Rethabile. The failing component is not in the network."

"No," said Ma-Rethabile Khumalo, who had known for a while.

“It’s this. It’s ours. We’re the thing that’s broken.”

She had the manifests with her because she has had a manifest with her every day since she was twenty-two.

Two hundred and forty-one crates, forty-one lots, and the lot codes fall into families, and some of the family codes carry a bar over them. She had asked about the bar in her first year and been told it was a shipping mark, and had believed it for six years, the way you believe the things you are told in your first year.

The unbarred families are components. She cuts them, she signs them, they go down the shaft on the carriers and into the galleries and get set in place, and their tolerance callouts revise slightly every consignment, and the guild calls the revisions *tolerance revisions*, and there is a standing item in the choir-line meeting about how annoying they are.

She laid the last eleven years of callouts out on the glass in her head, which she could do, because she is the person who signs them.

They revise in one direction. They revise at a hundred and eighty parts in a billion a year.

Twenty-seven years of shimming. Every component cut to the pitch the node *has*, not the pitch the node *was*. Not repair. Not once. Chasing a falling number down with new metal every twelve months so the thing does not tear itself apart on the way down, and calling it maintenance in the manifest heading because the word in the manifest heading was translated in 2073 by a committee.

“Hayibo,” she said, to nobody, quietly.

Then the barred families.

The barred ones are not dimensioned. That is what she had never once noticed, in six years, holding them in her two hands: their callouts are not in micrometres, they are in frequency, and they are specified to nine digits, and there is no function on this planet that needs nine digits. You do not build a part to nine digits. You build a *standard* to nine digits.

She did not have to look up the number. She had it under her thumb,

on the glass, twice: 312.442 and change, the sealed value, the sleep value, the value the port side of the peg had not held since the year of the Flare.

The barred lots are cut to the sleep. To what it was. To nine places.

And they leave. Every consignment, the barred families are shipped out and never appear in any installation record, and she had assumed they went into galleries she did not have access to, because there are always galleries you do not have access to.

She said it out loud once, badly, and then again.

"They're not buying parts. They're buying the fork."

Tefo said, "Say it in a sentence I can carry to the Elder."

"This node kept the calibration. All of it slept and ours slept longest and cleanest, and when everything else in the network drifted or died there was one true pitch left and it was under Johannesburg." She had her hand flat on the glass. It was cool. In a mountain at thirty-eight degrees, the glass is cool. "Sixty metres of it. That's the whole of the world's standard, right there, and it doesn't have a certificate and it doesn't need one. The consignments do two jobs. The unbarred ones keep this one running at whatever pitch it has this year, which is a losing job and they have never said otherwise. The barred ones carry the true number out. Copies of it. Reference pitch, going out to wherever the rest of the thing is, in gold, because gold is the only thing that will hold a number that hard for that long."

"And when this one has drifted far enough that the sealed length is the only true thing left?"

"Then the sealed length is sixty metres of glass in a hole in a country, and there is nobody in the network who can come down here and hold it." She looked up at him. "That is the sentence."

Ma-Rethabile took her the last two hundred metres, to the threshold, and did not talk on the way, and the reason she did not talk became apparent when they got there, which is that there is nothing to say about it that survives being in the room.

The wall with the names is in the antechamber. Fifty-one minutes'

worth of custodians, June 2064, and the ones who came down and did not sing but stood, and their names are on it too, in the same hand, at the same size.

The threshold itself is dust and worked stone and a floor that has been swept by feet in one narrow track.

“In 2074 the party asked to stand here,” Ma-Rethabile said. “The Accord permitted it. There were eleven of them and they were here for six hours and I was twenty-three years old and I held the lamp.” She put the stick against the wall. “Nothing woke. Nothing hummed, nothing warmed, nothing moved. They were courteous. They packed their instruments and they thanked the order in the protocol form and they have never asked again, and it is not in any published record, and I am telling you and I am not going to argue about what it means.”

“They can’t hold it.”

“I said what I saw.”

“Ma-Rethabile, they can’t hold it. That’s why the amendment—” She stopped. She got the folded amendment out of her bag, the working text, five days old, which she had read twice for the tonnage and once for the seventy-third clause because everyone reads the seventy-third clause.

Annex C. Twenty-two pages of tables. Jigs. Blanks. Die sets, with the master forms enumerated. A schedule of instruction, eleven years, with host-nation personnel required present at threshold on a rising count: two, then six, then fourteen, then forty. Payment in technique, and the technique itemised, and the itemised technique is not materials science and is not hull physics. It is how to cut a standard. It is how to hold a number.

Nobody had read Annex C. It is tables. She had not read Annex C, and she is the one who would have understood it.

“The volume amendment isn’t an offtake deal,” she said. “It’s a hand-over. Four hundred and sixty tonnes and eleven years and forty of our people standing where we are standing, and at the end of it there is no counterparty in the loop and the machine is ours to keep in tune, and it will need keeping in tune forever, and forever is not a figure of speech in this room.”

Ma-Rethabile Khumalo picked up her stick.

"I halted two consignments," she said. "In '84 and in '91. Both times because the barred lots came back with the number changed in the ninth place and I could not find out who had changed it or why, and I could not explain that to a commission without explaining what the ninth place was, and I had promised my predecessor I would not. So I halted them and said nothing and let the world call it superstition." She started up the track. "I have been asked whether we can trust them for twenty-seven years, ausi, and it is not the question. The question is the one you have just read in a table. Will we take it back. I have never been able to answer it and neither has my order and neither, I think, has yours."

Naledi stayed a moment longer with the lamp.

Then she went up, three thousand two hundred metres of cage, forty minutes, and came out into a light that hurt, with no certificate, no gallery access, no instruments in her own name, four days, and an arithmetic on the back of a photostat of her grandmother's handwriting that could not be given to a committee to hold for her.

She found a phone that worked and asked the port operator for Berth 7C.



Klaus

Custos, non Conditor